

SQANTI3 report

Unique Genes: 1386

Unique Isoforms: 2428

Transcript Classification

Category	Isoforms, count
FSM	2428
ISM	0
NIC	0
NNC	0
Genic Genomic	0
Antisense	0
Fusion	0
Intergenic	0
Genic Intron	0

Gene Classification

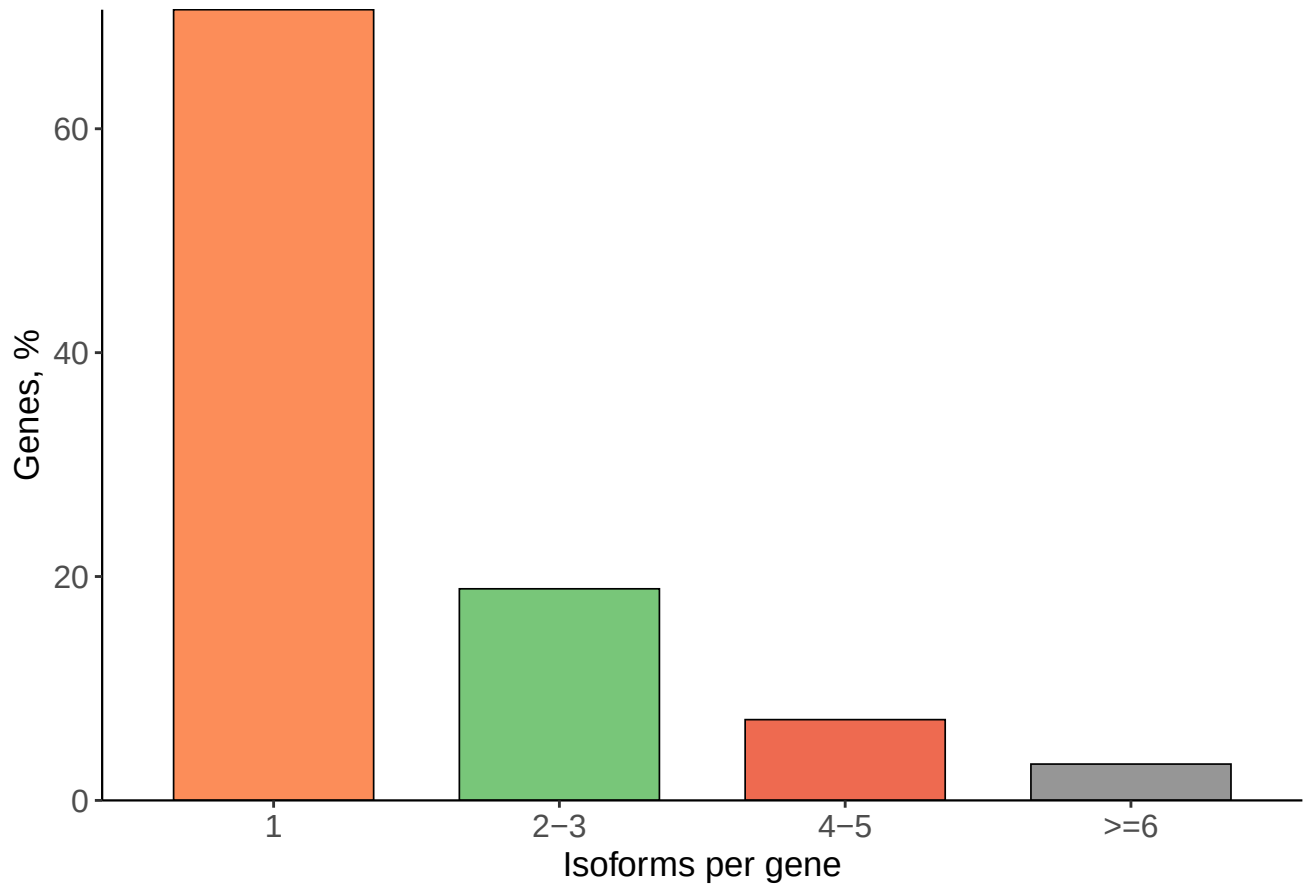
Category	Genes, count
Annotated Genes	1386
Novel Genes	0

Splice Junction Classification

Category	SJs, count	Percent
Known canonical	6025	97.21
Known Non-canonical	173	2.79
Novel canonical	0	0.00
Novel Non-canonical	0	0.00

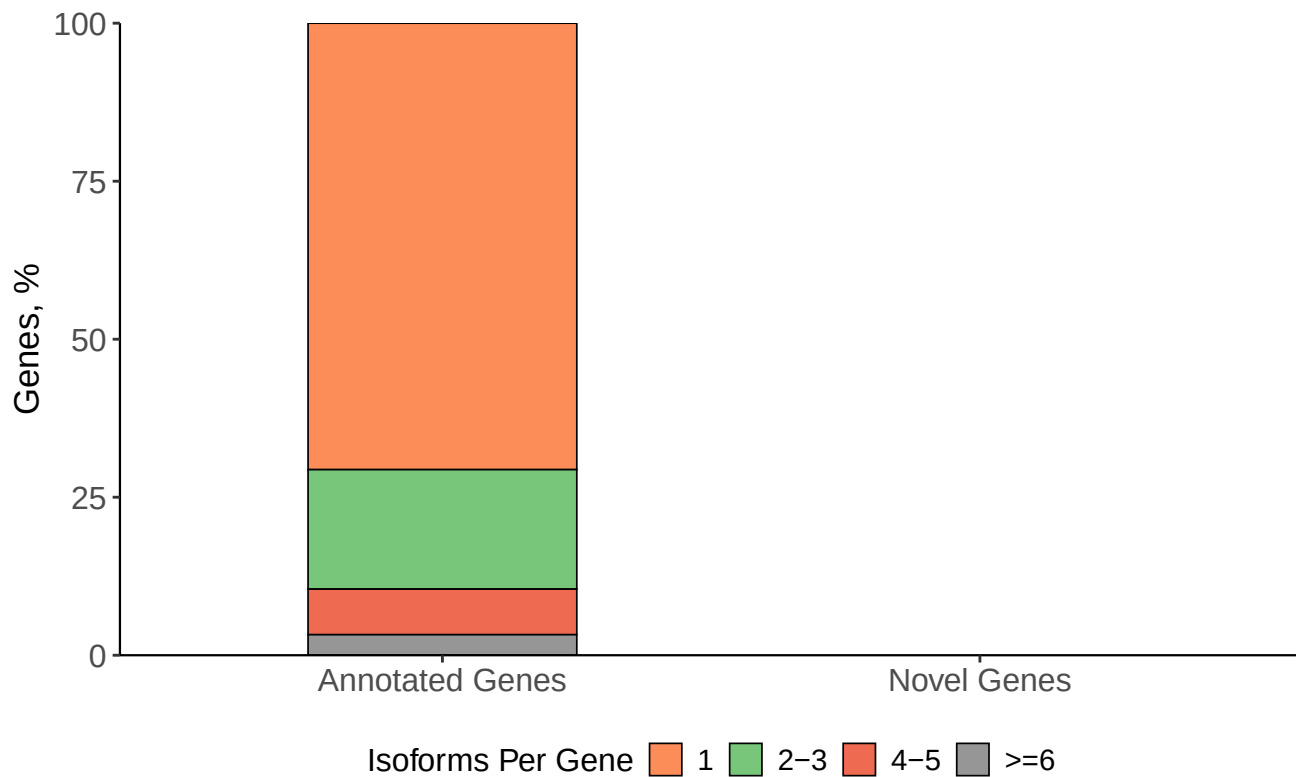
Gene Characterization

Number of Isoforms per Gene



Number of Isoforms per Gene

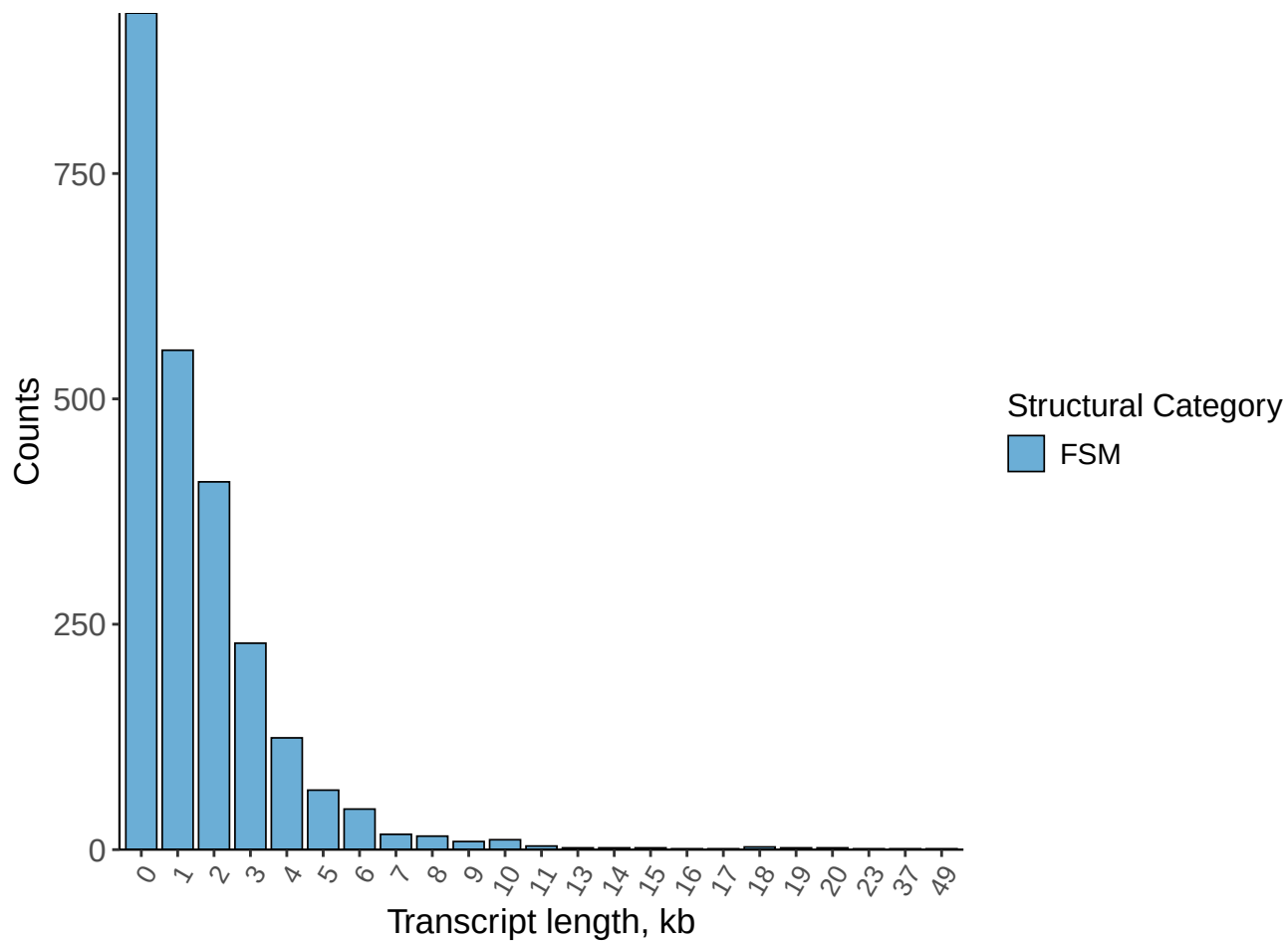
Known vs Novel Genes



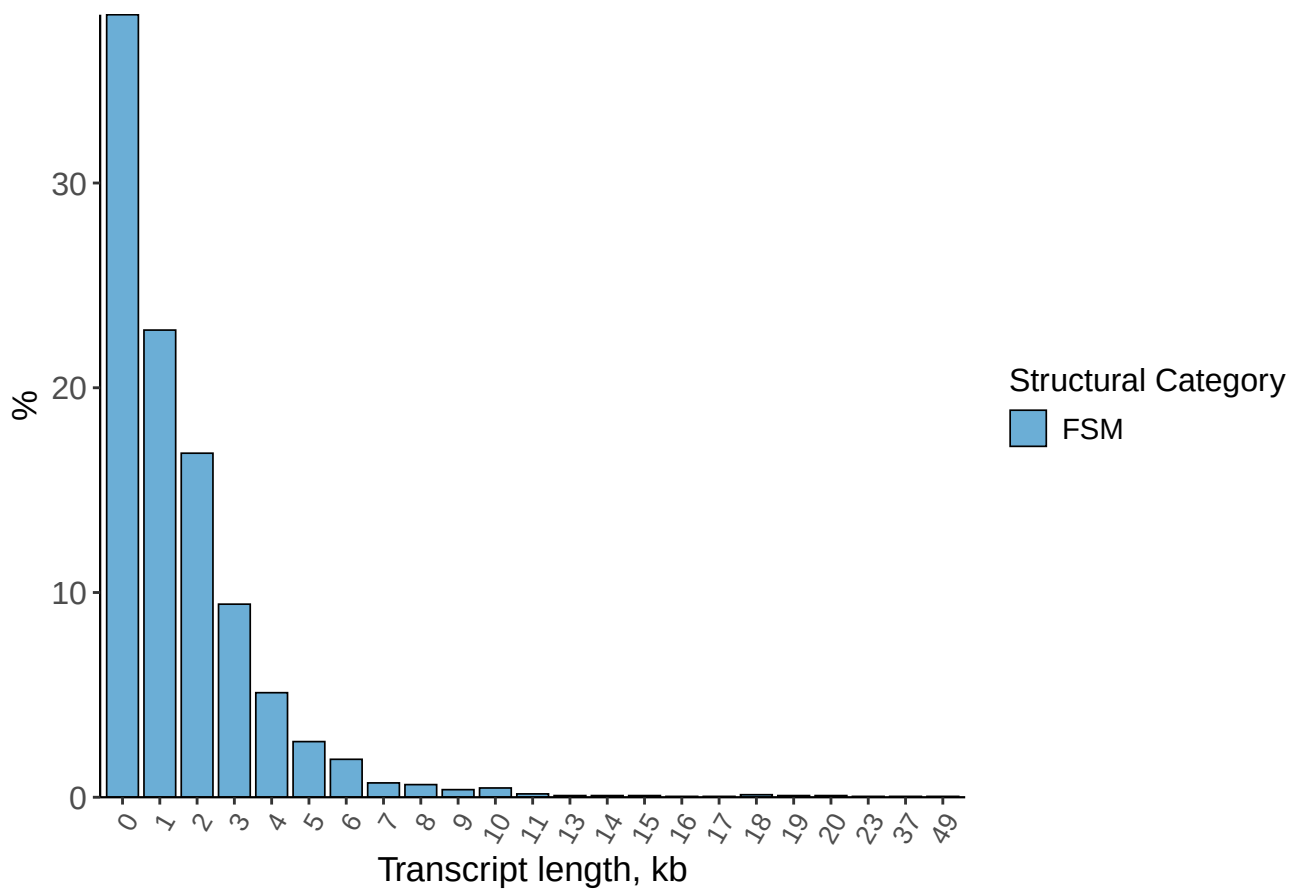
Distribution of Mono- vs Multi-Exon Transcripts



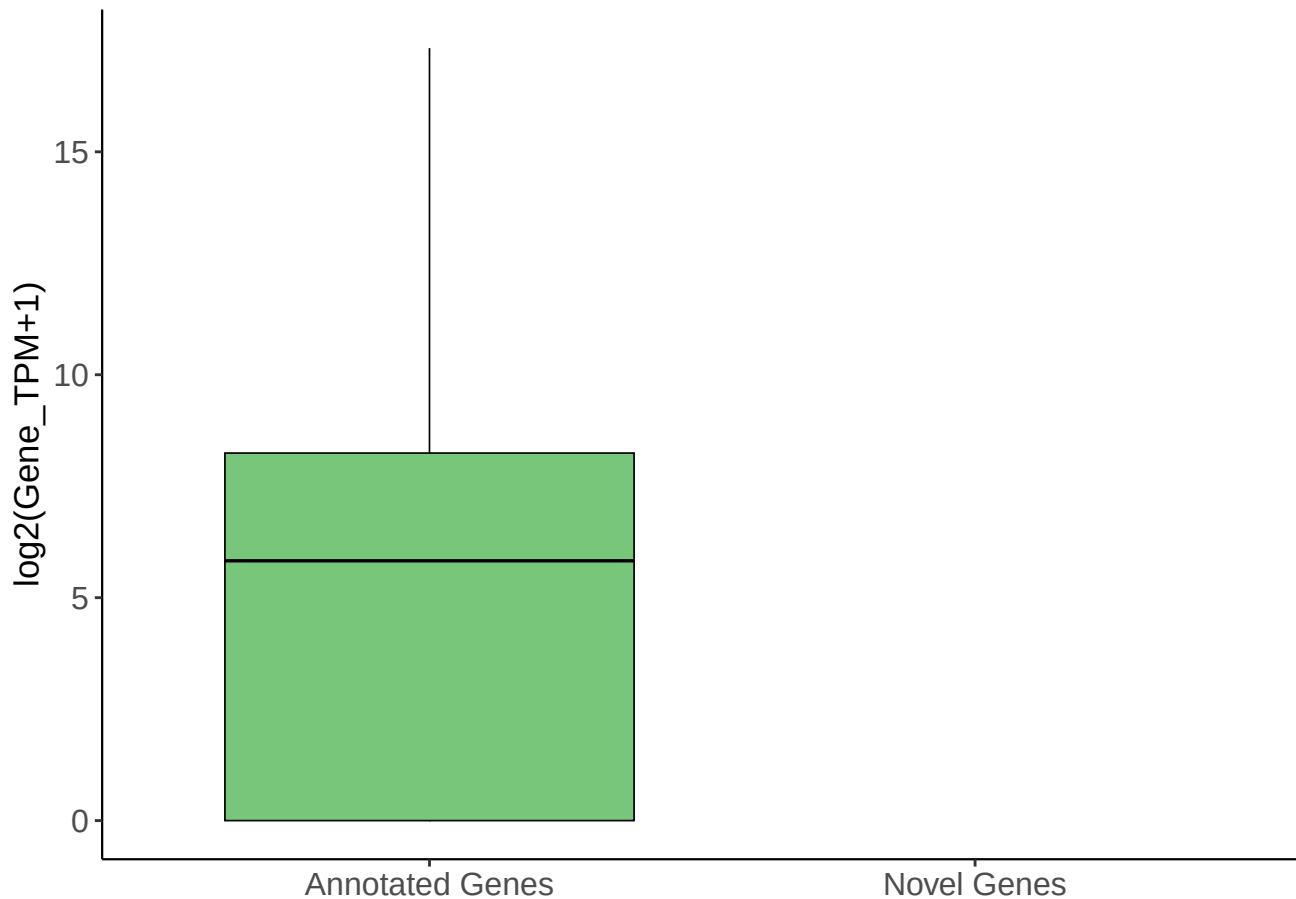
Structural Categories by Transcript Length



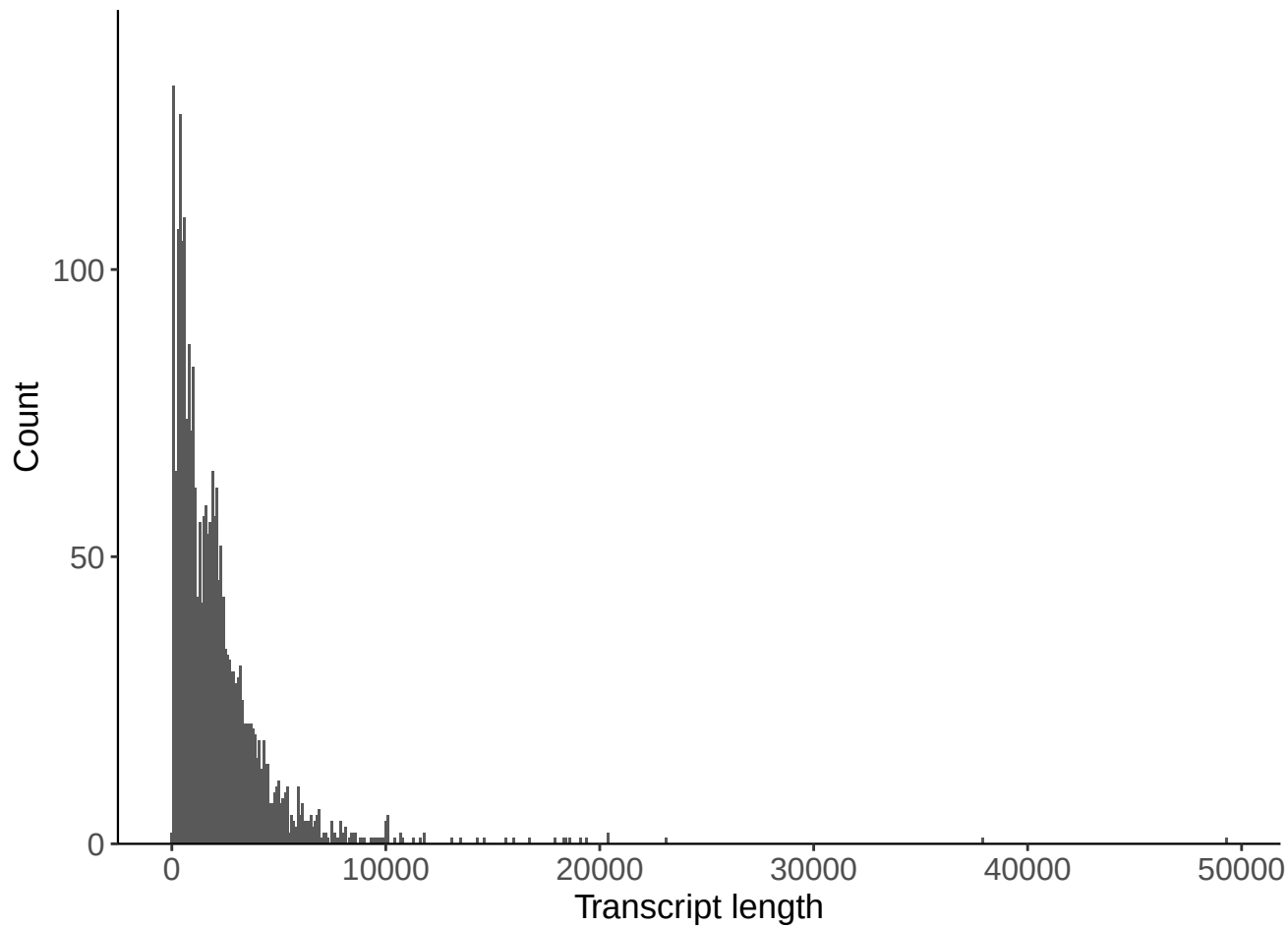
Structural Categories by Transcript Length



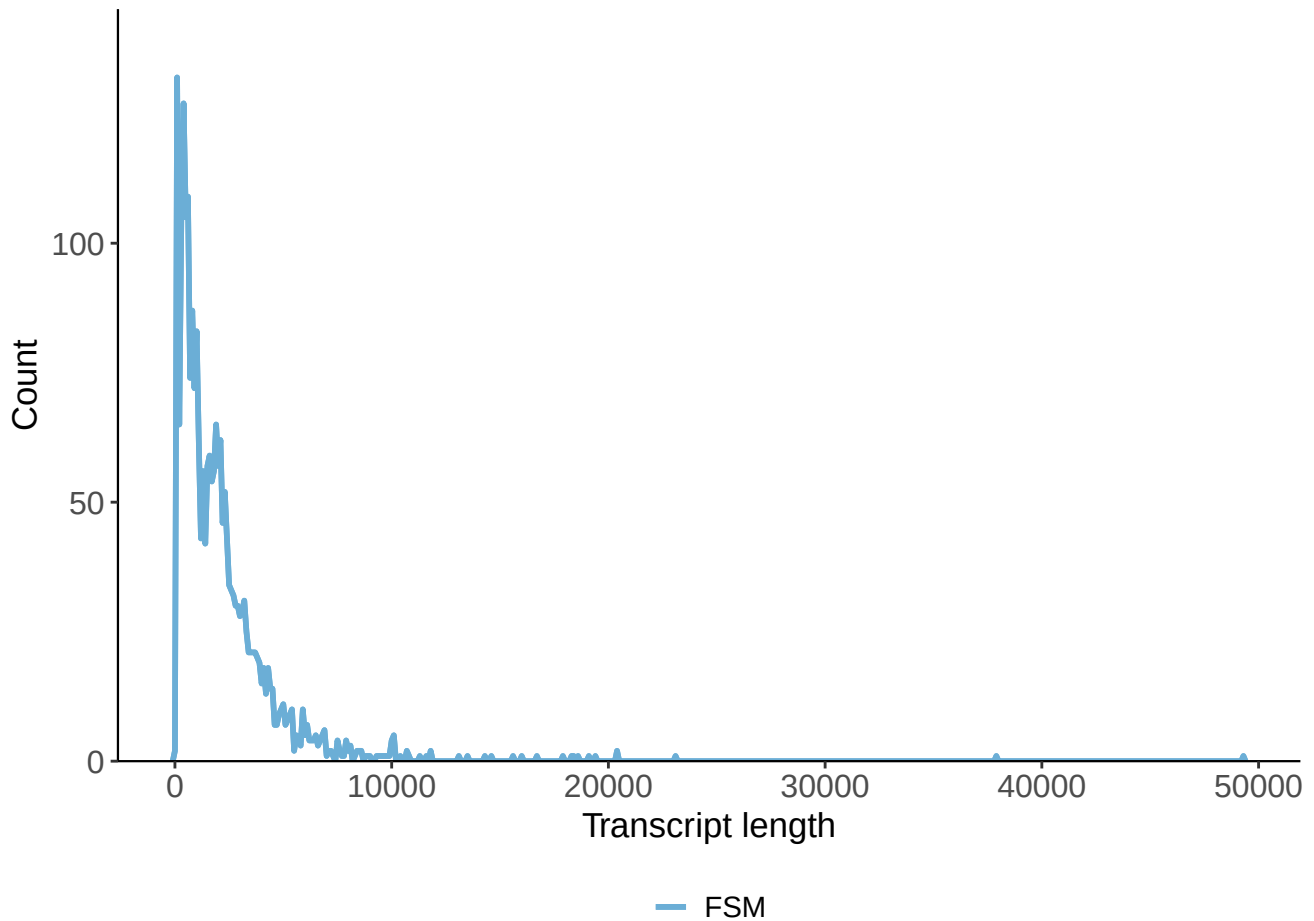
Annotated vs Novel Gene Expression



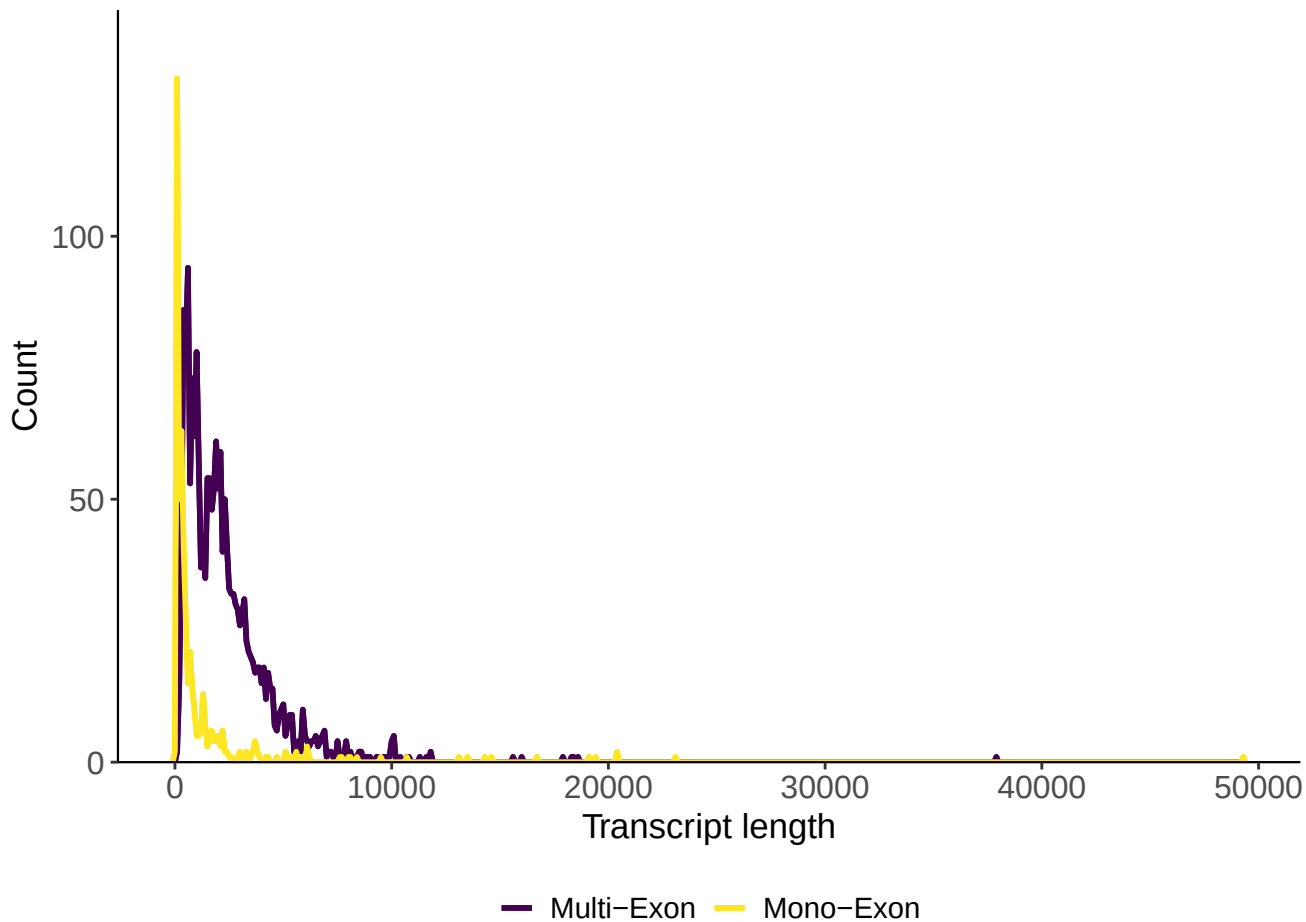
All Transcript Lengths Distribution



Transcript Lengths Distribution by Structural Category

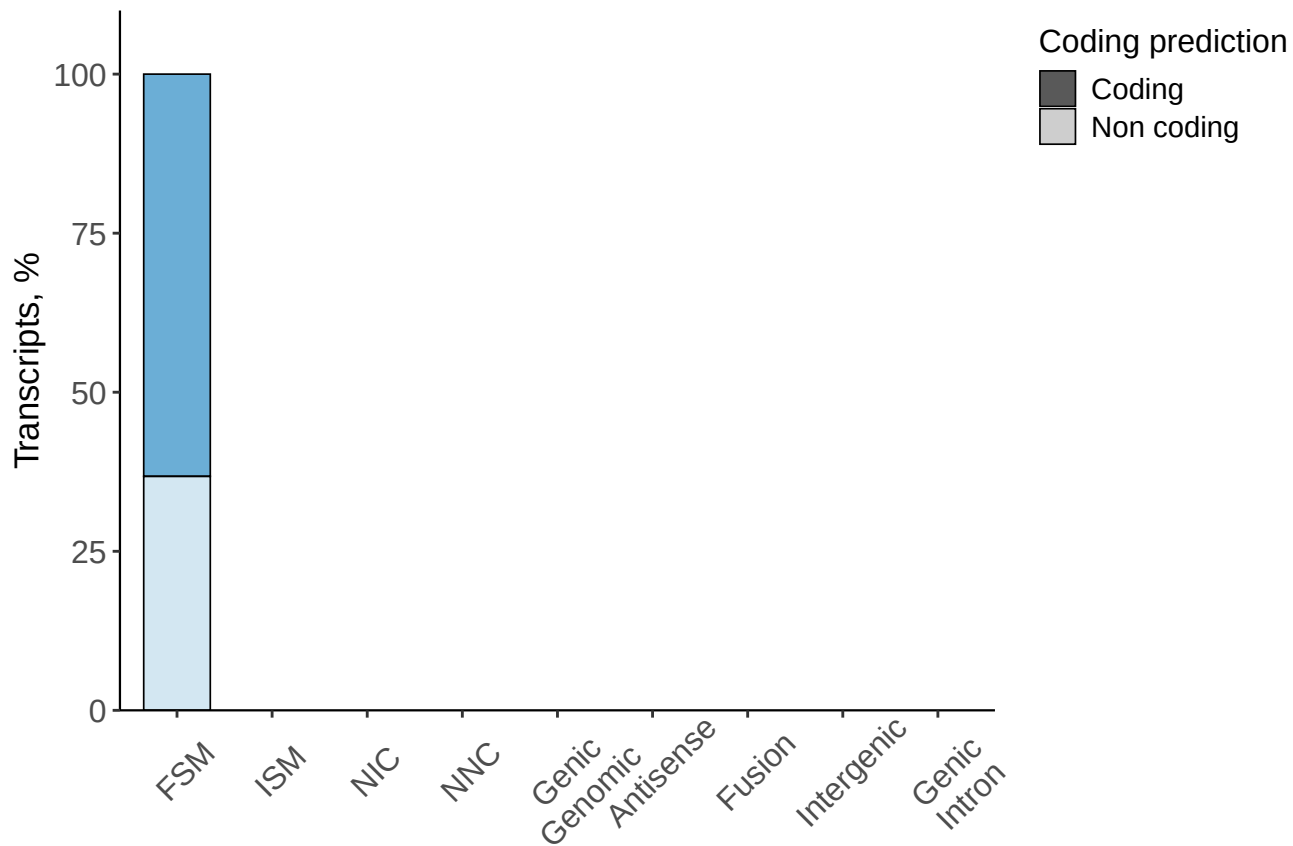


Mono- vs Multi- Exon Transcript Lengths Distribution

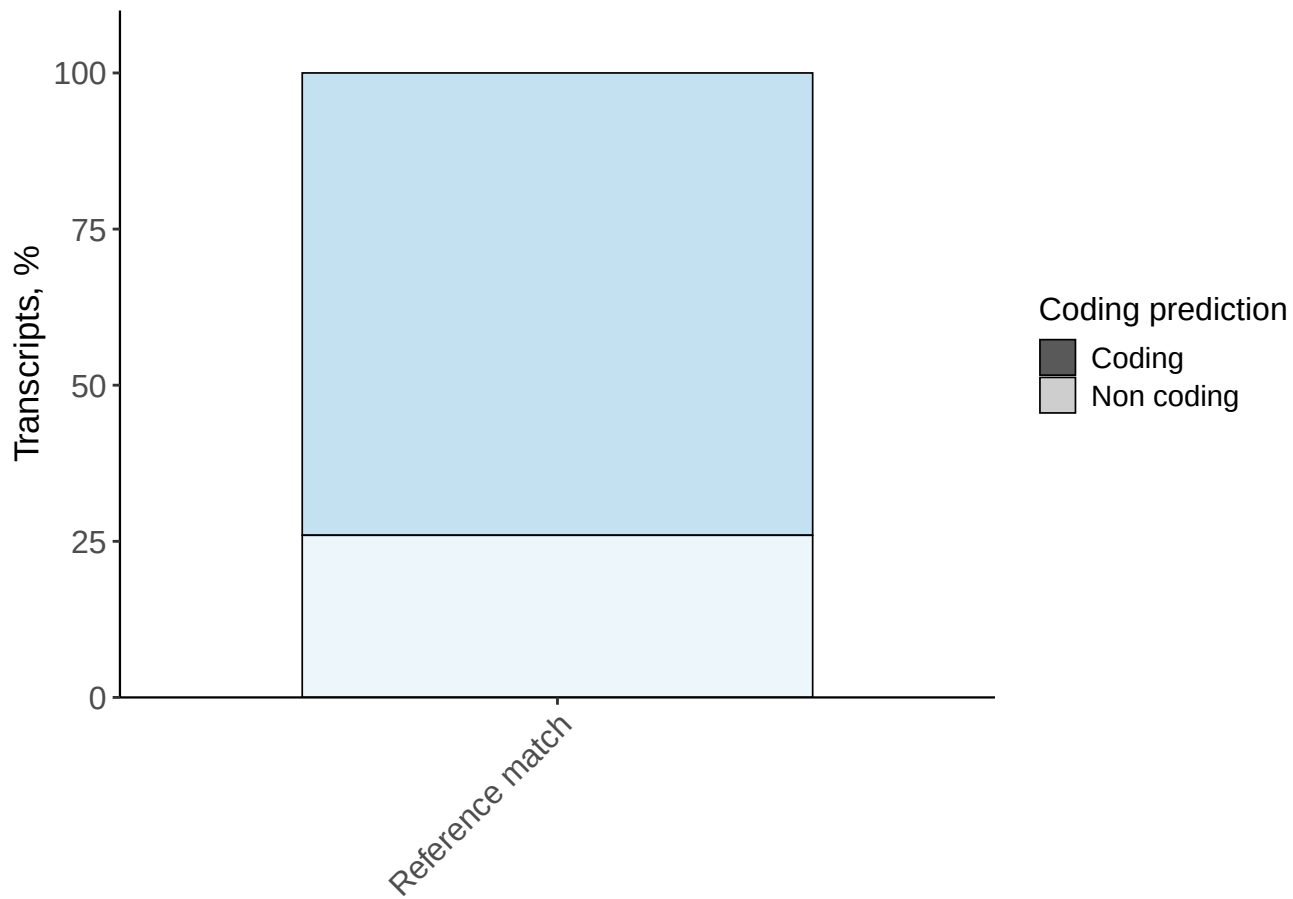


Structural Isoform Characterization

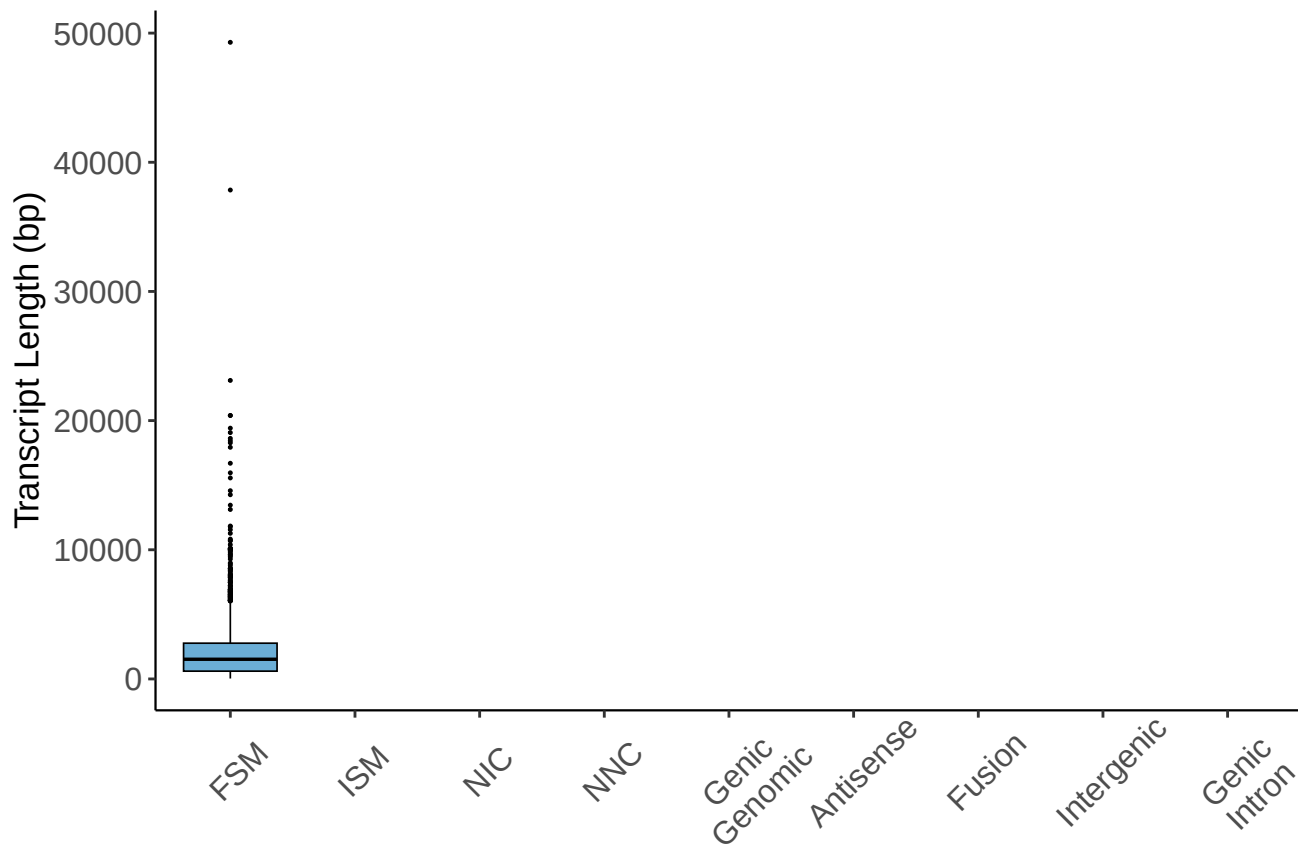
Isoform Distribution Across Structural Categories



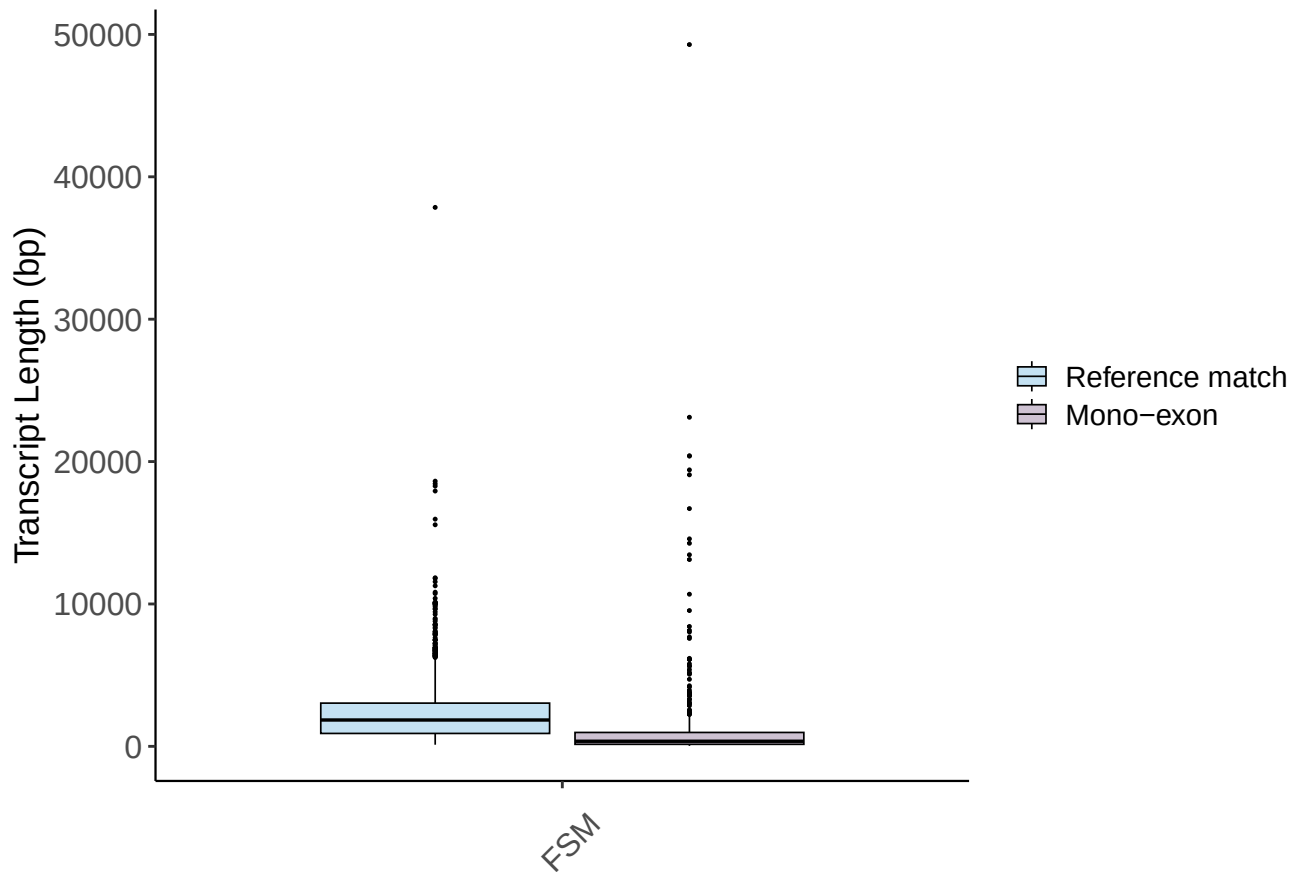
Isoform Distribution Across FSM



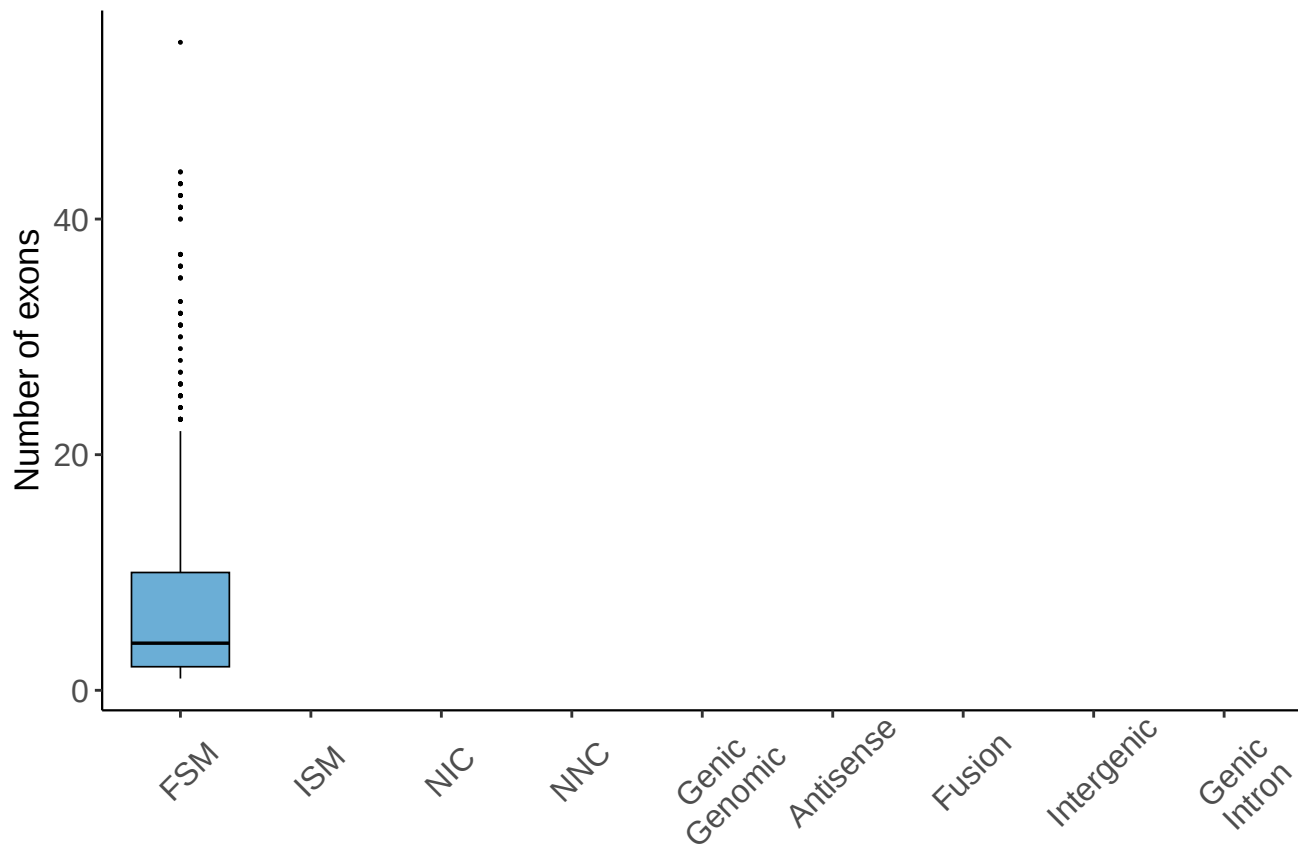
Transcript Lengths by Structural Classification



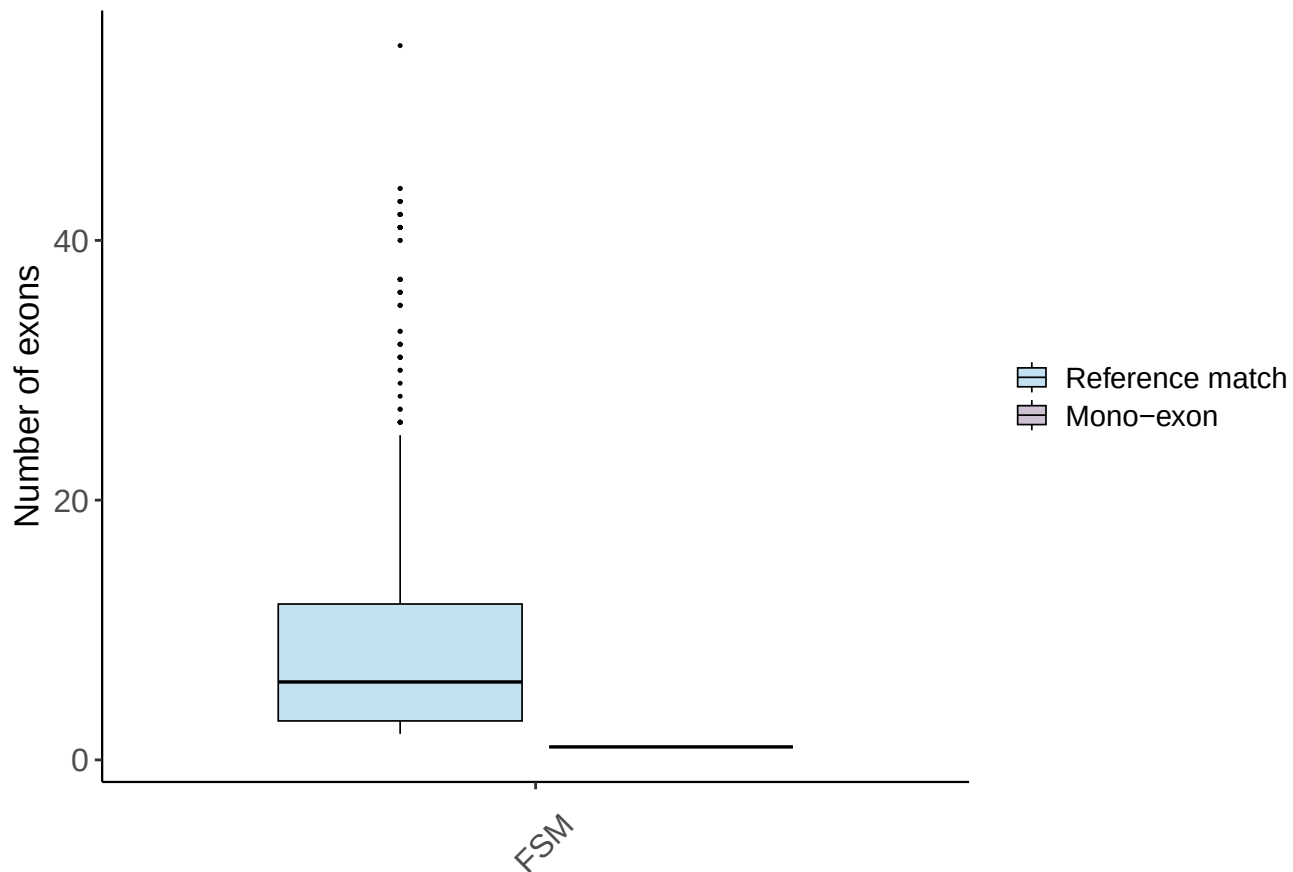
Transcript Lengths by Subcategory



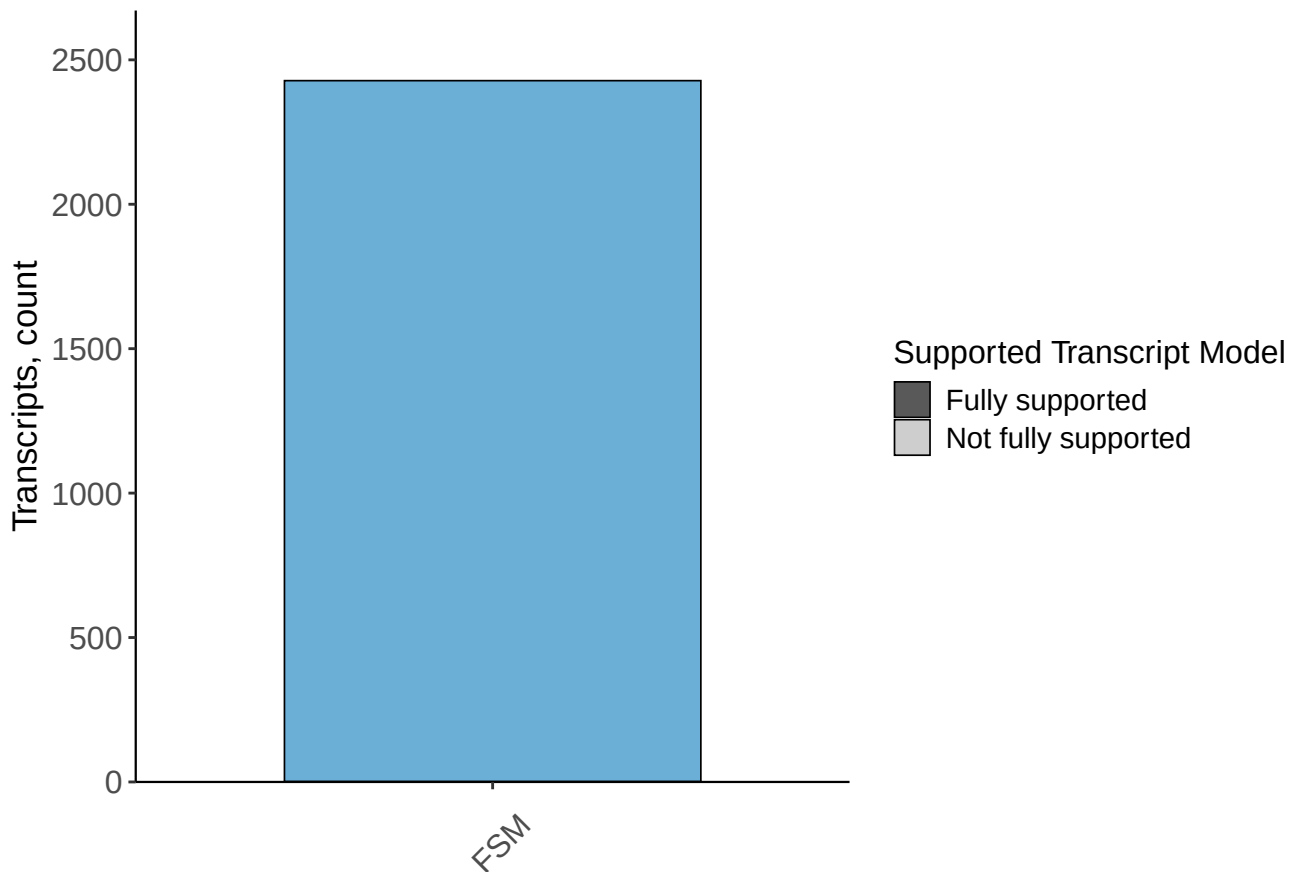
Exon Counts by Structural Classification



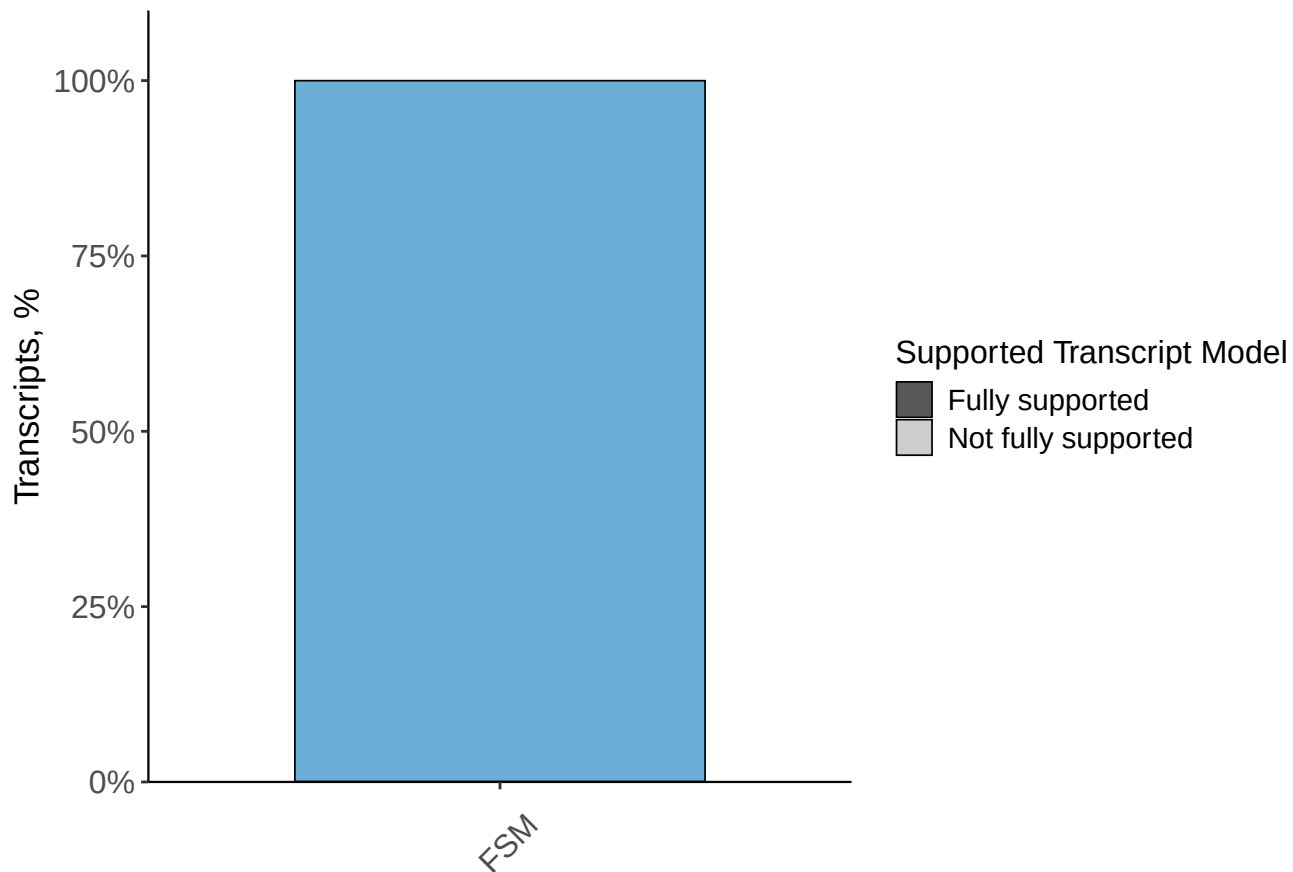
Exon Counts by Subcategory



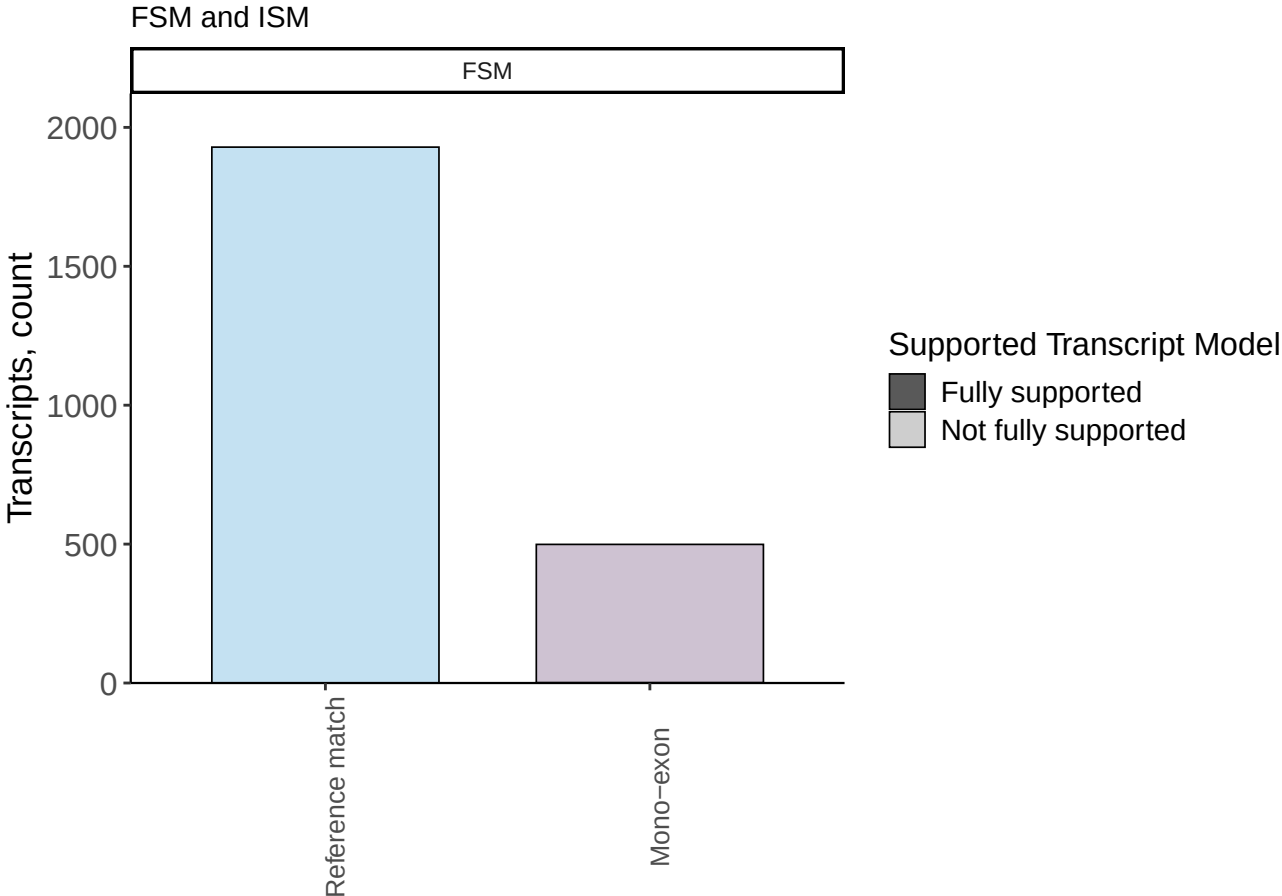
Isoform Distribution Across Structural Categories



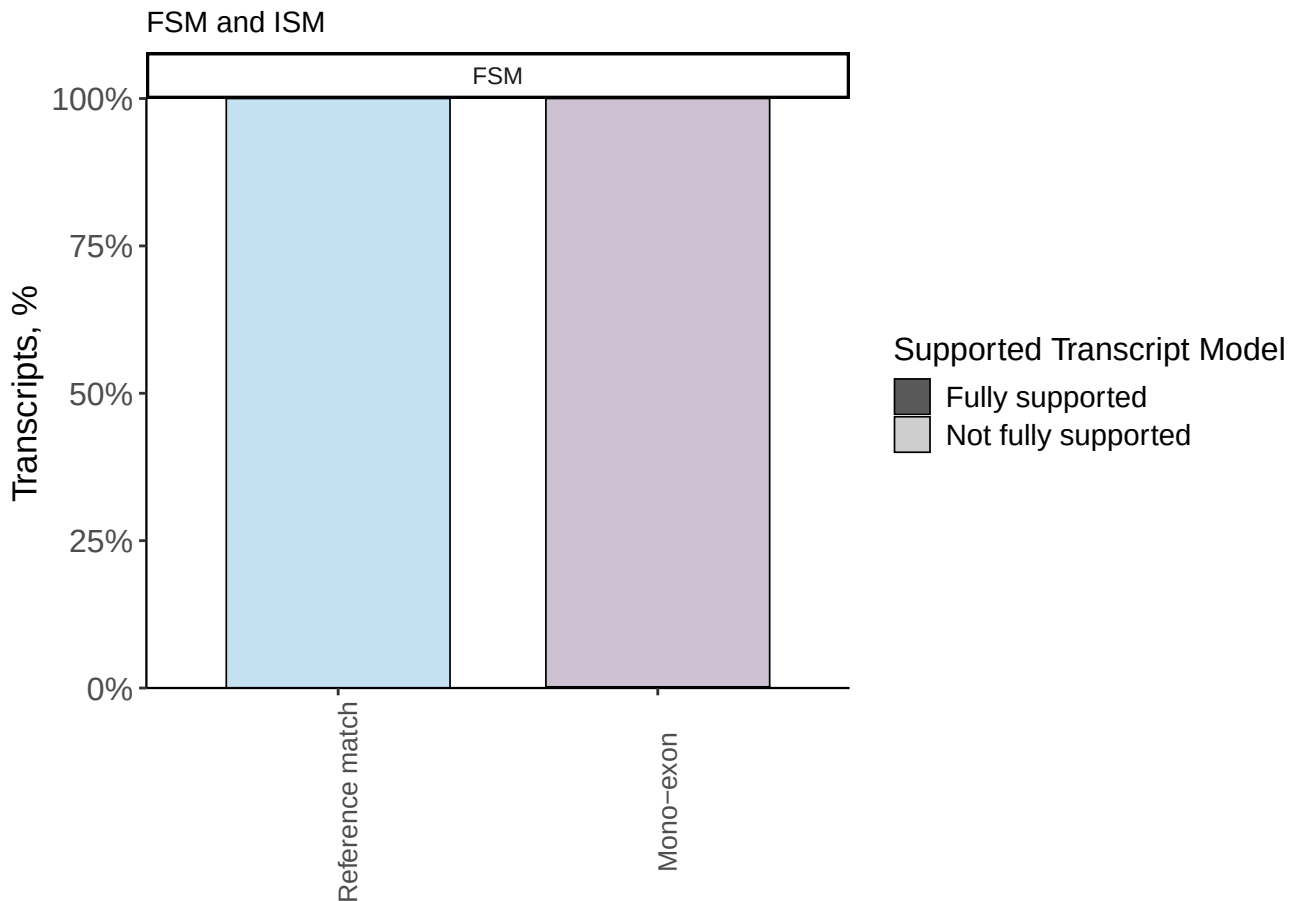
Isoform Distribution Across Structural Categories



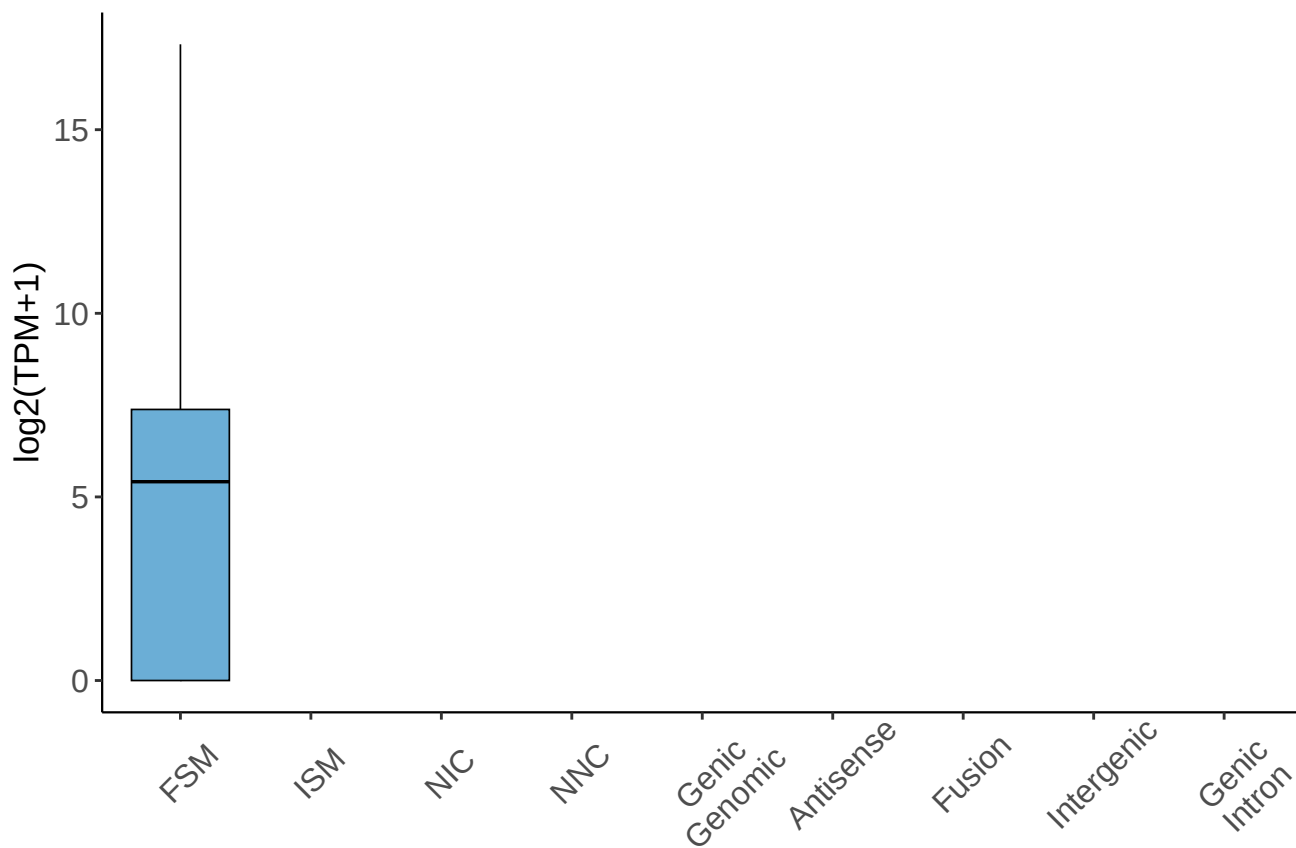
Isoform Distribution Across Structural Subcategories



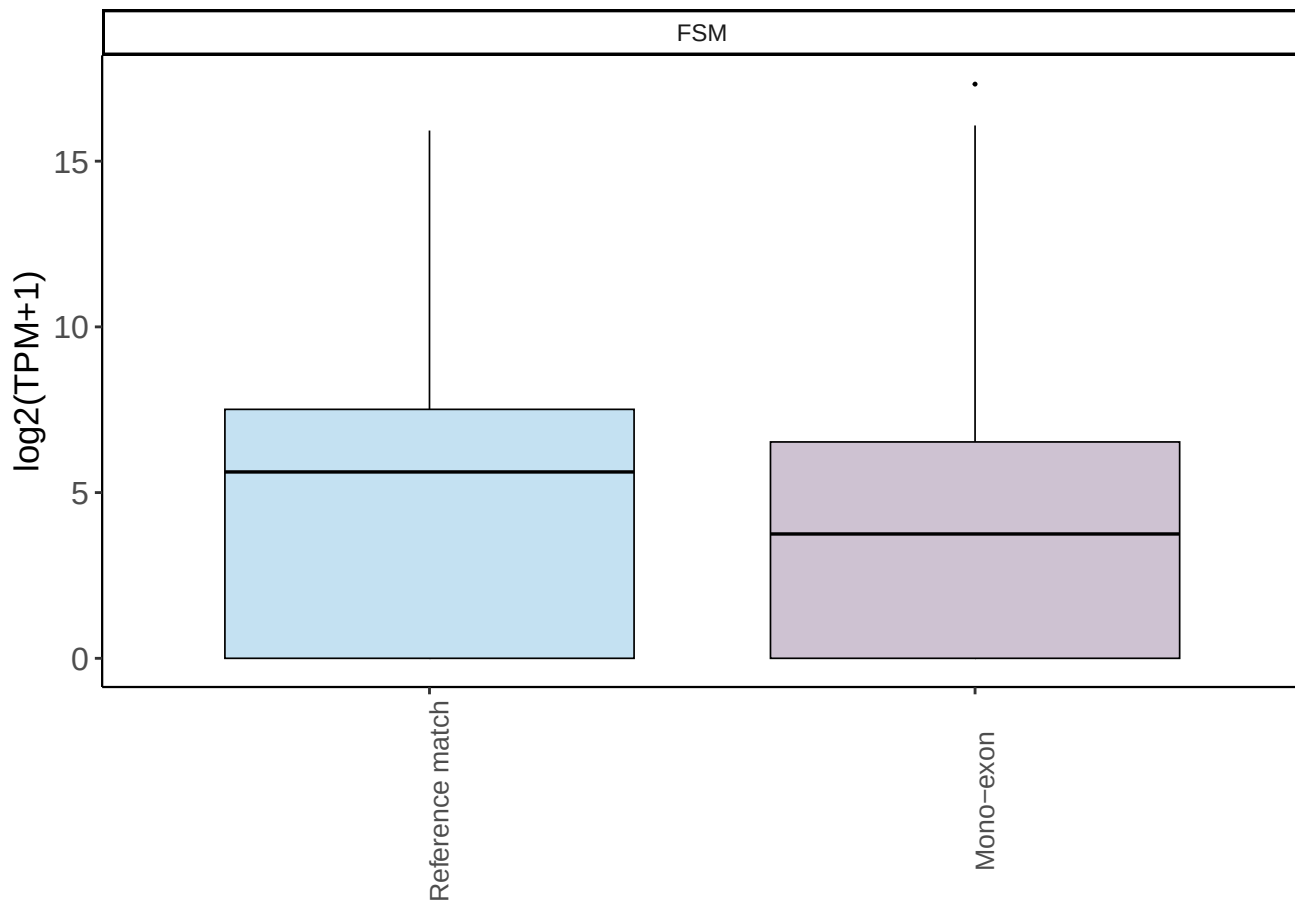
Isoform Distribution Across Structural Subcategories



Transcript Expression by Structural Category

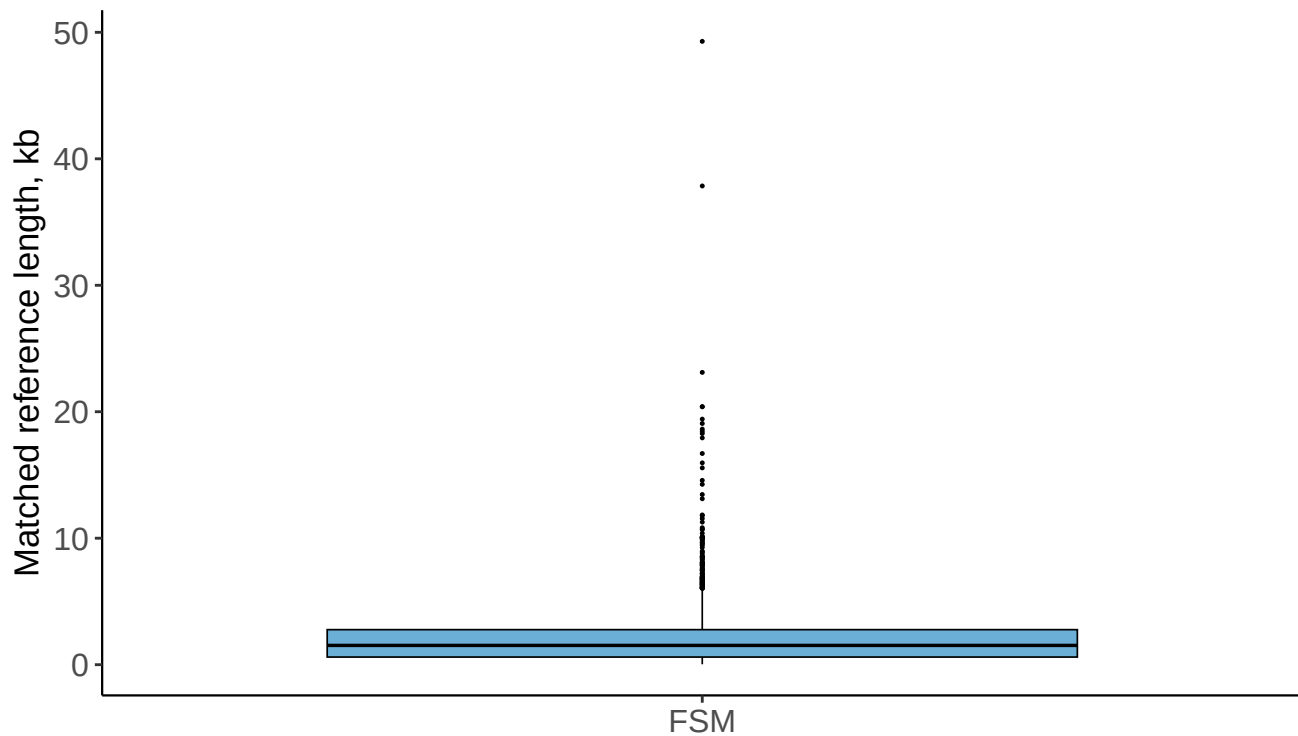


Transcript Expression by Subcategory



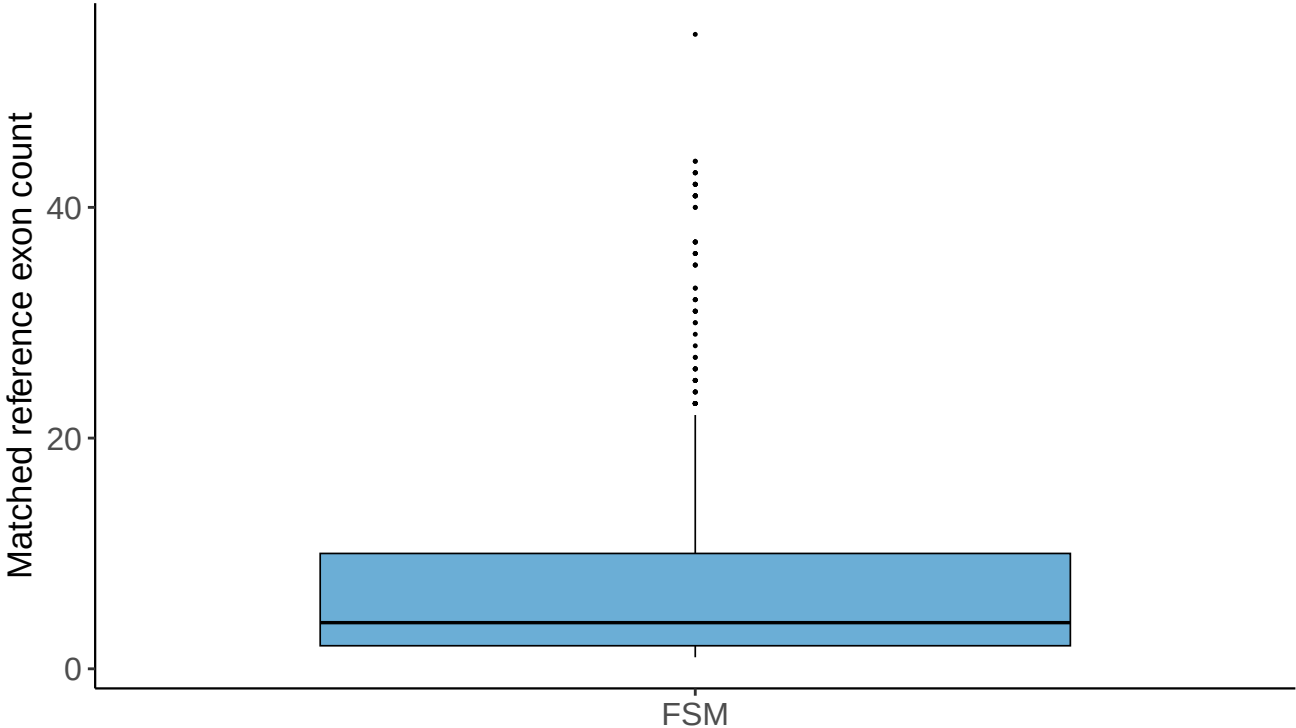
Length Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories



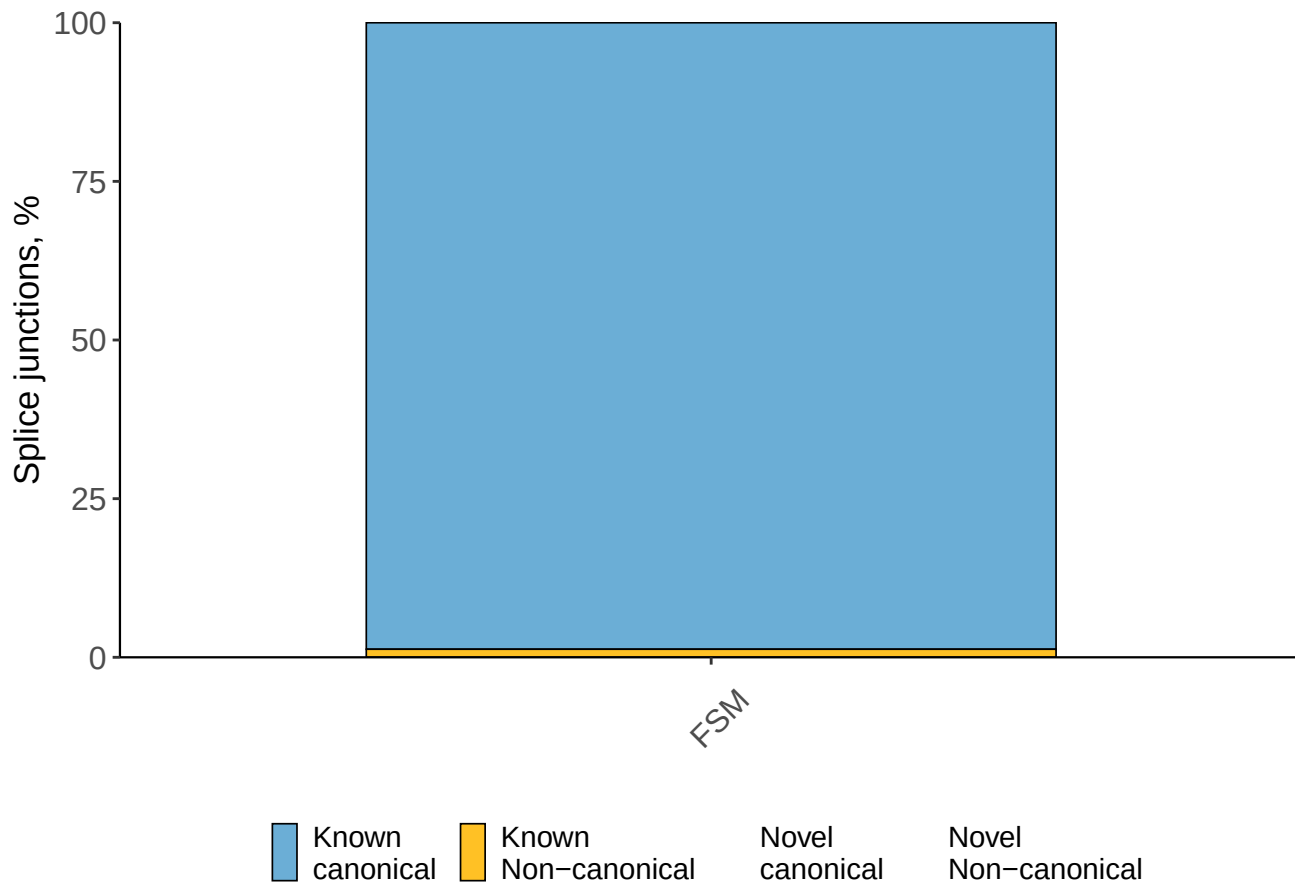
Exon Count Distribution of Matched Reference Transcripts

Applicable Only to FSM and ISM Categories

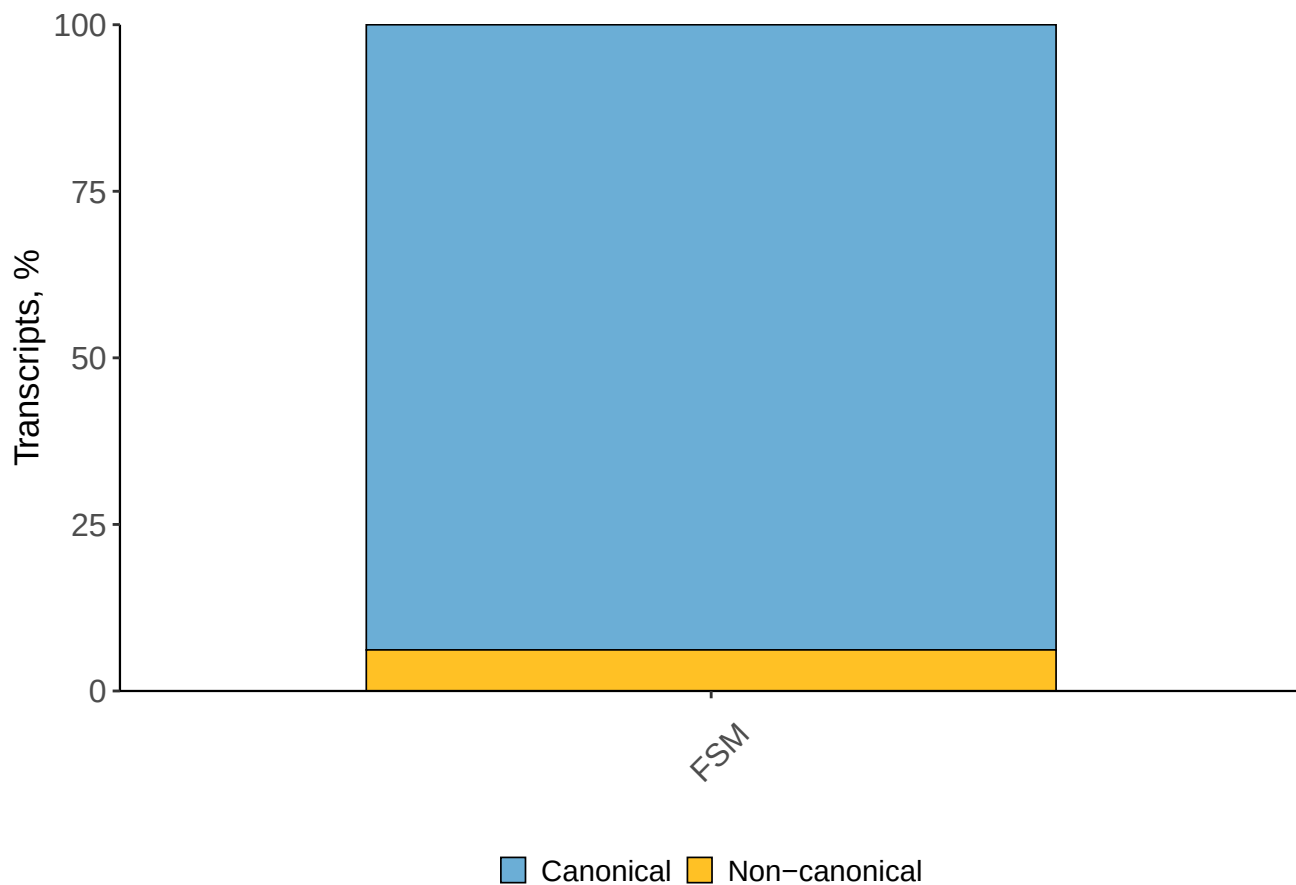


Splice Junction Characterization

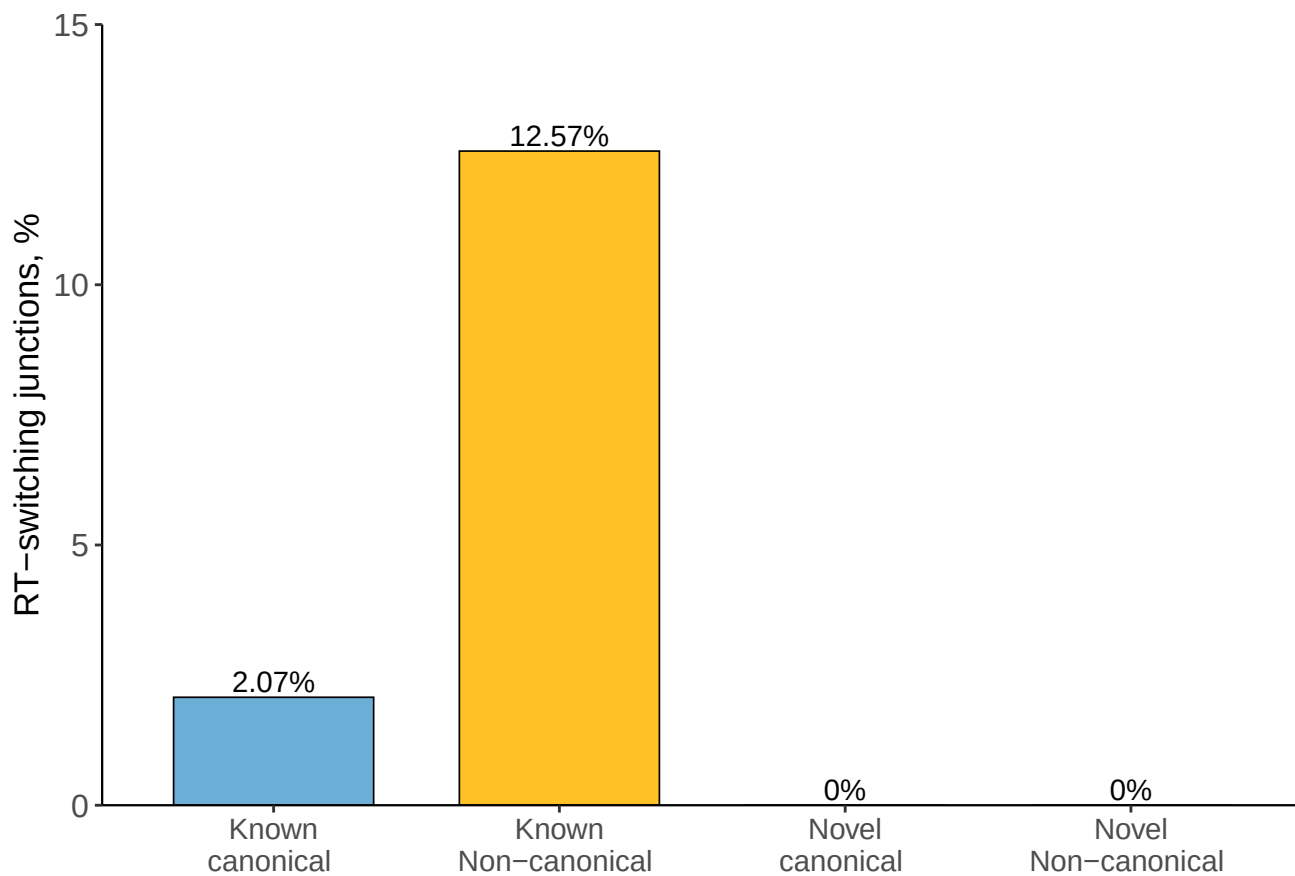
Distribution of Splice Junctions by Structural Classification



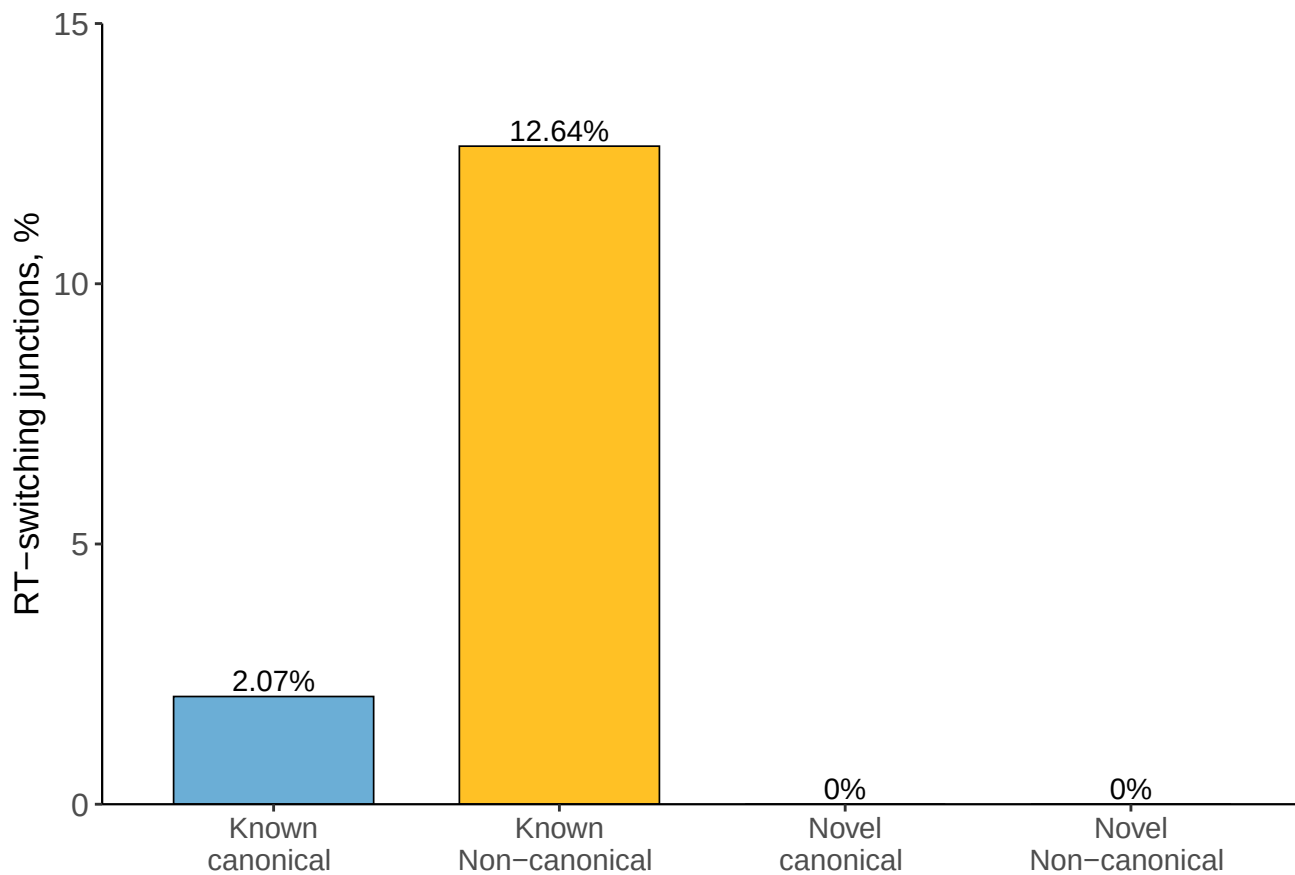
Distribution of Transcripts by Splice Junctions



RT-Switching All Junctions



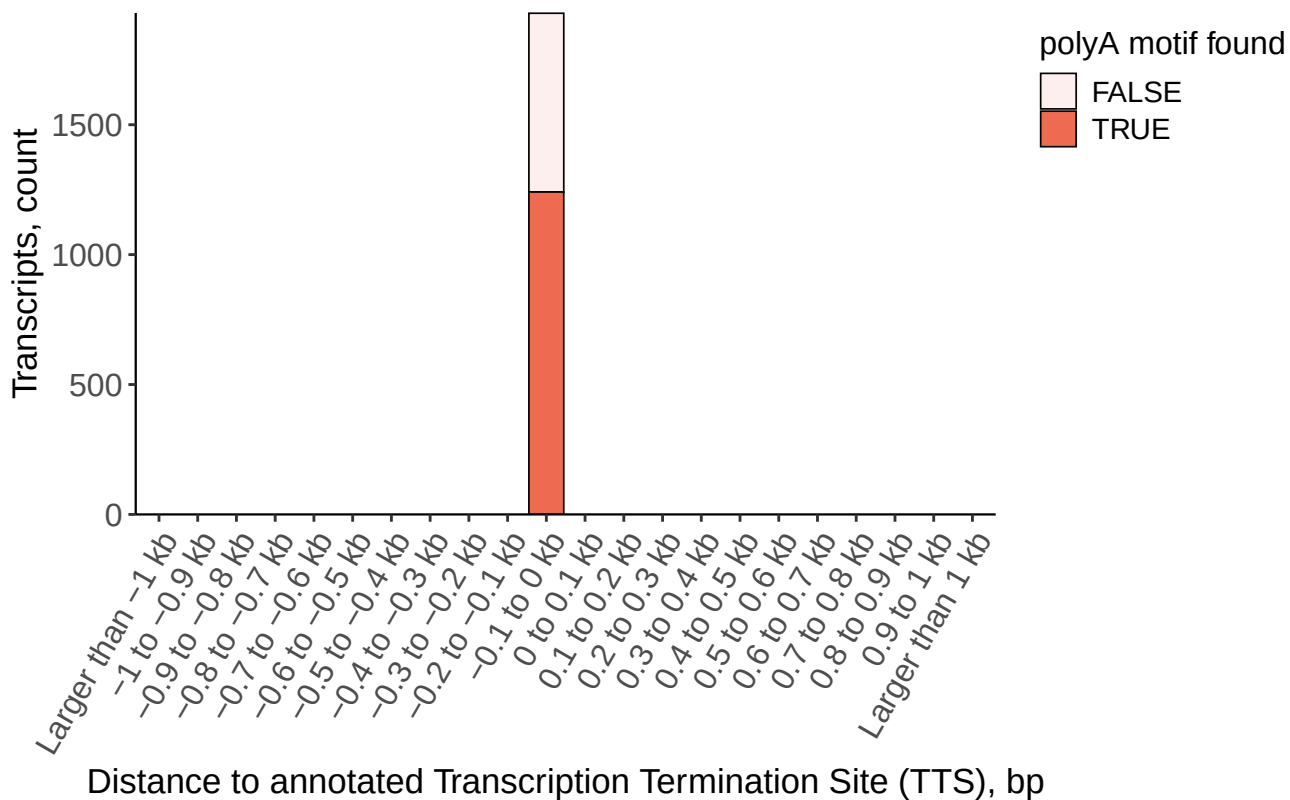
Unique Junctions RT-switching



Comparison With Annotated TSS and TTS

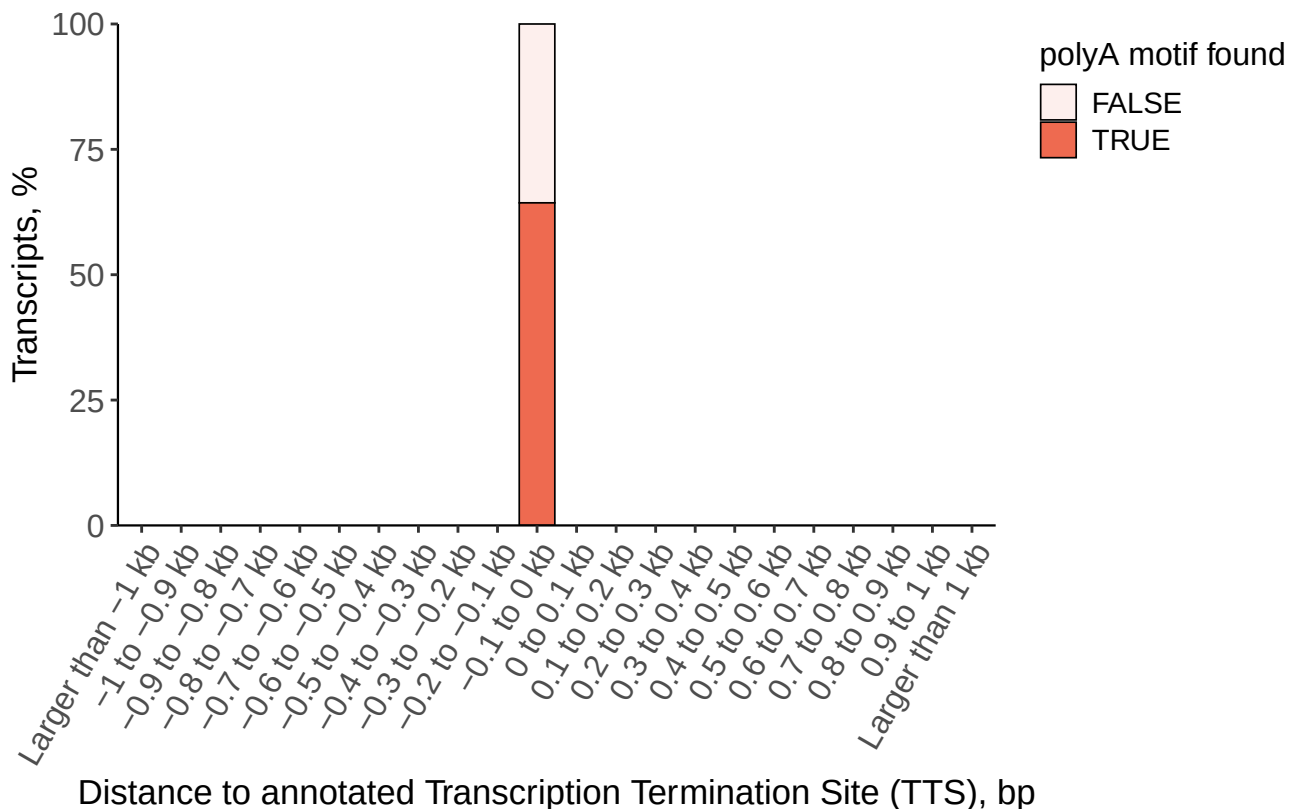
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



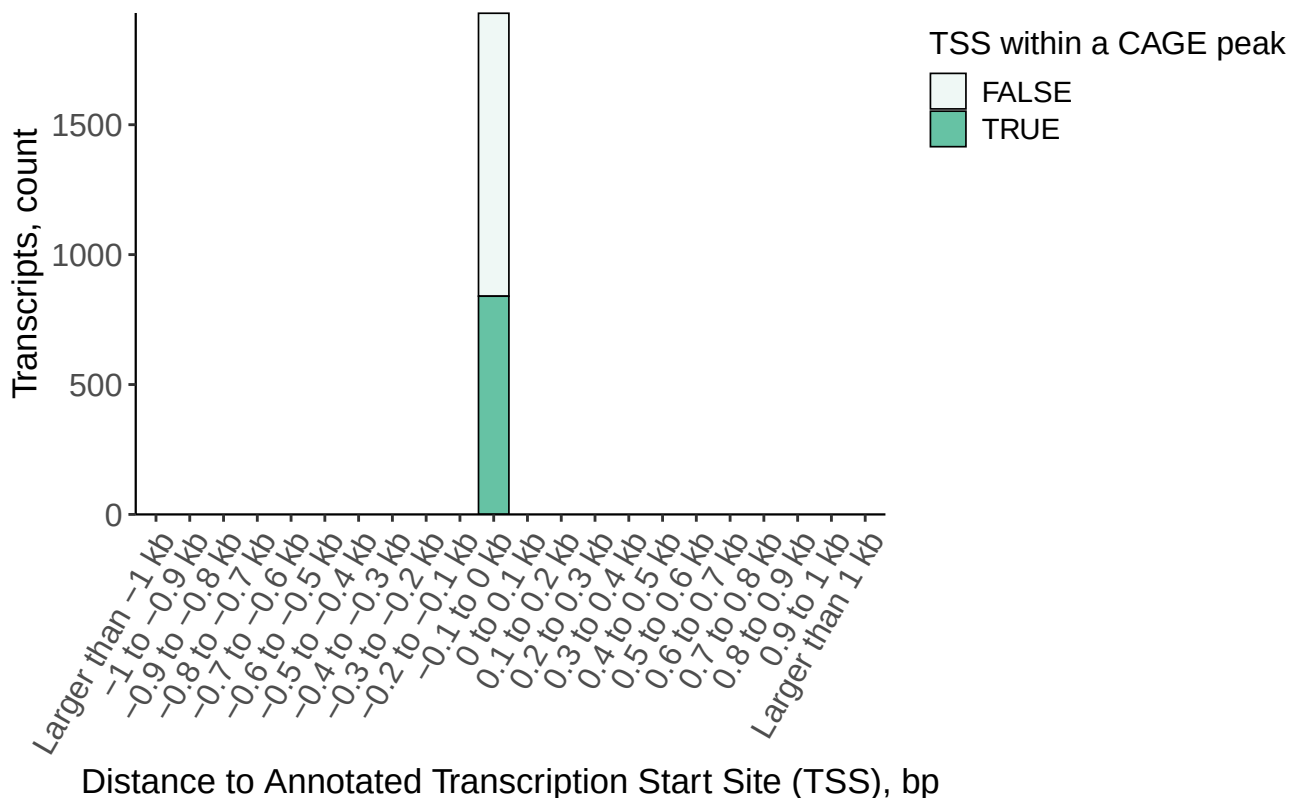
Distance to annotated Transcription Termination Site (TTS) FSM

Negative values indicate upstream of annotated termination site



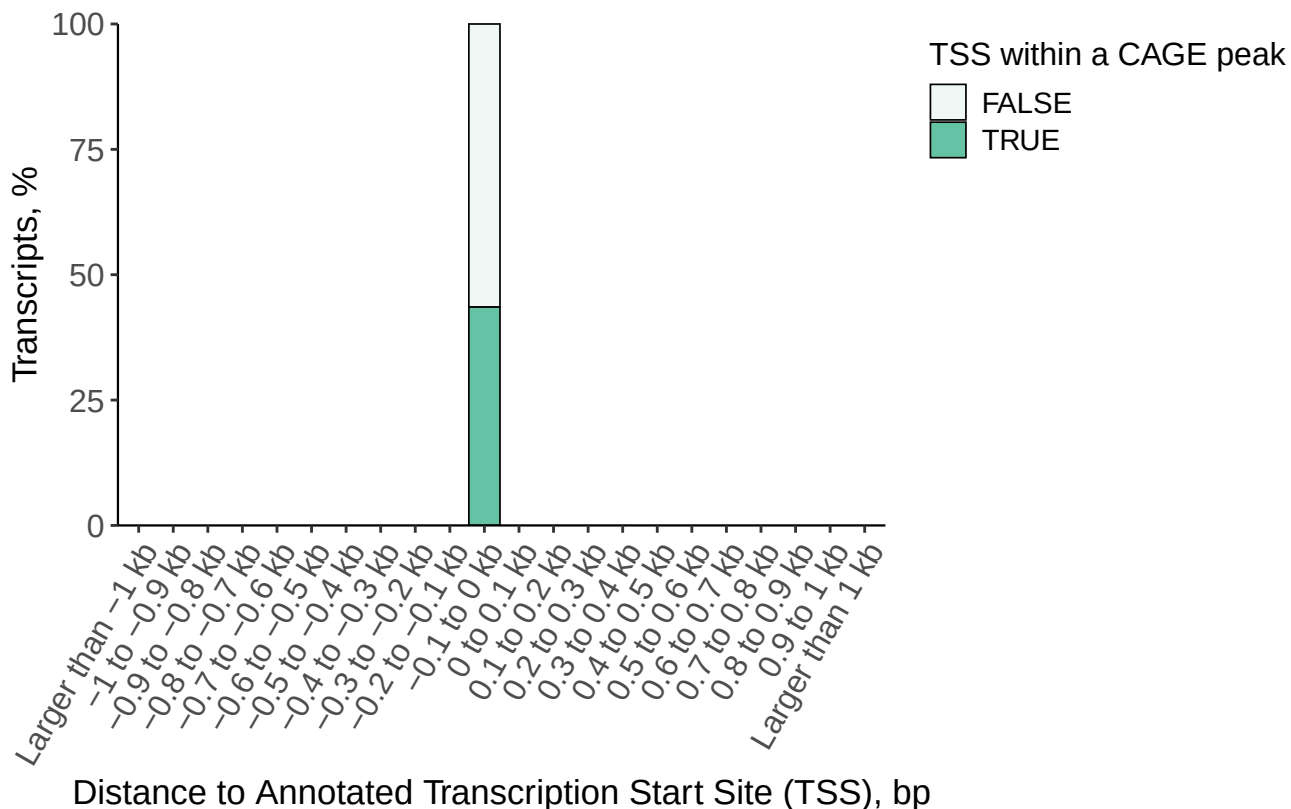
Distance to Annotated Transcription Start Site for FSM

Negative values indicate downstream of annotated TSS



Distance to Annotated Transcription Start Site for FSM

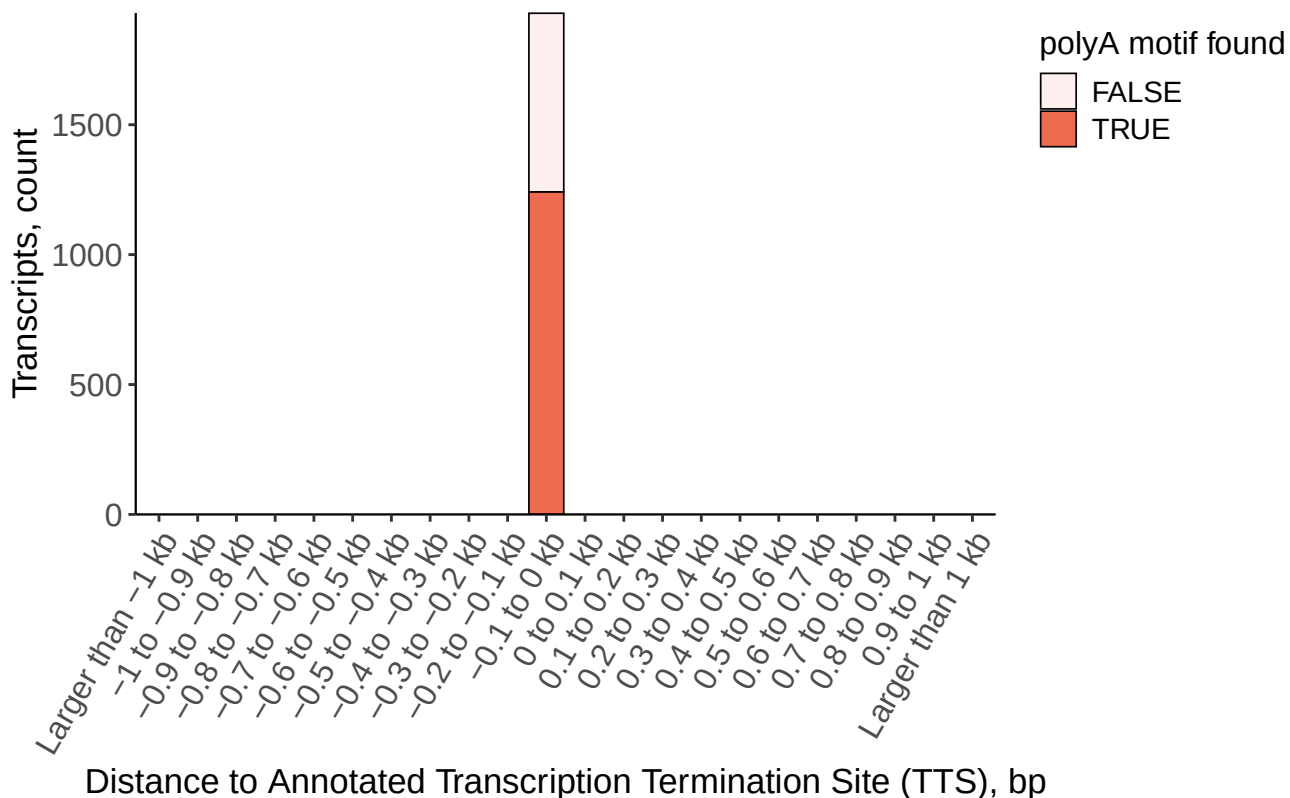
Negative values indicate downstream of annotated TSS



*Comparison With Annotated TSS and TTS
by Subcategories*

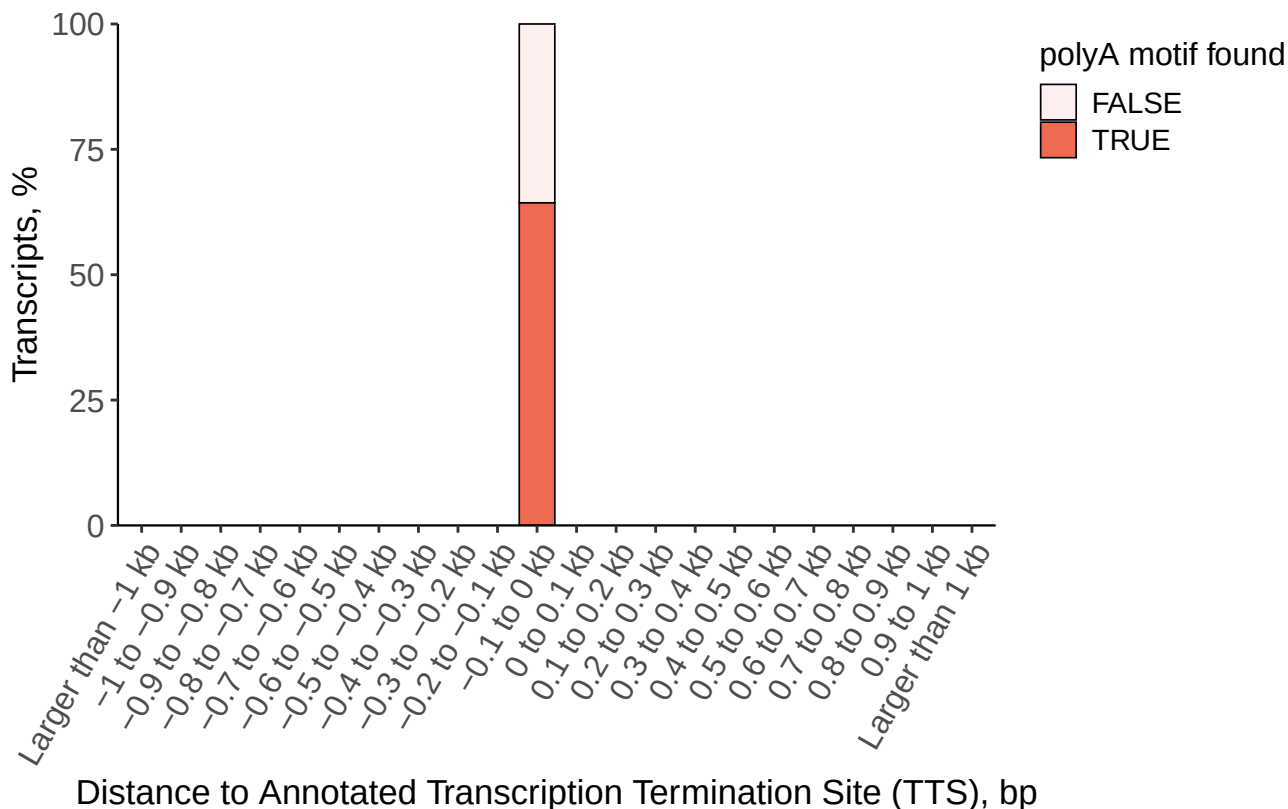
Distance to annotated Transcription Termination Site (TTS) FSM Reference Match

Negative values indicate upstream of annotated termination site



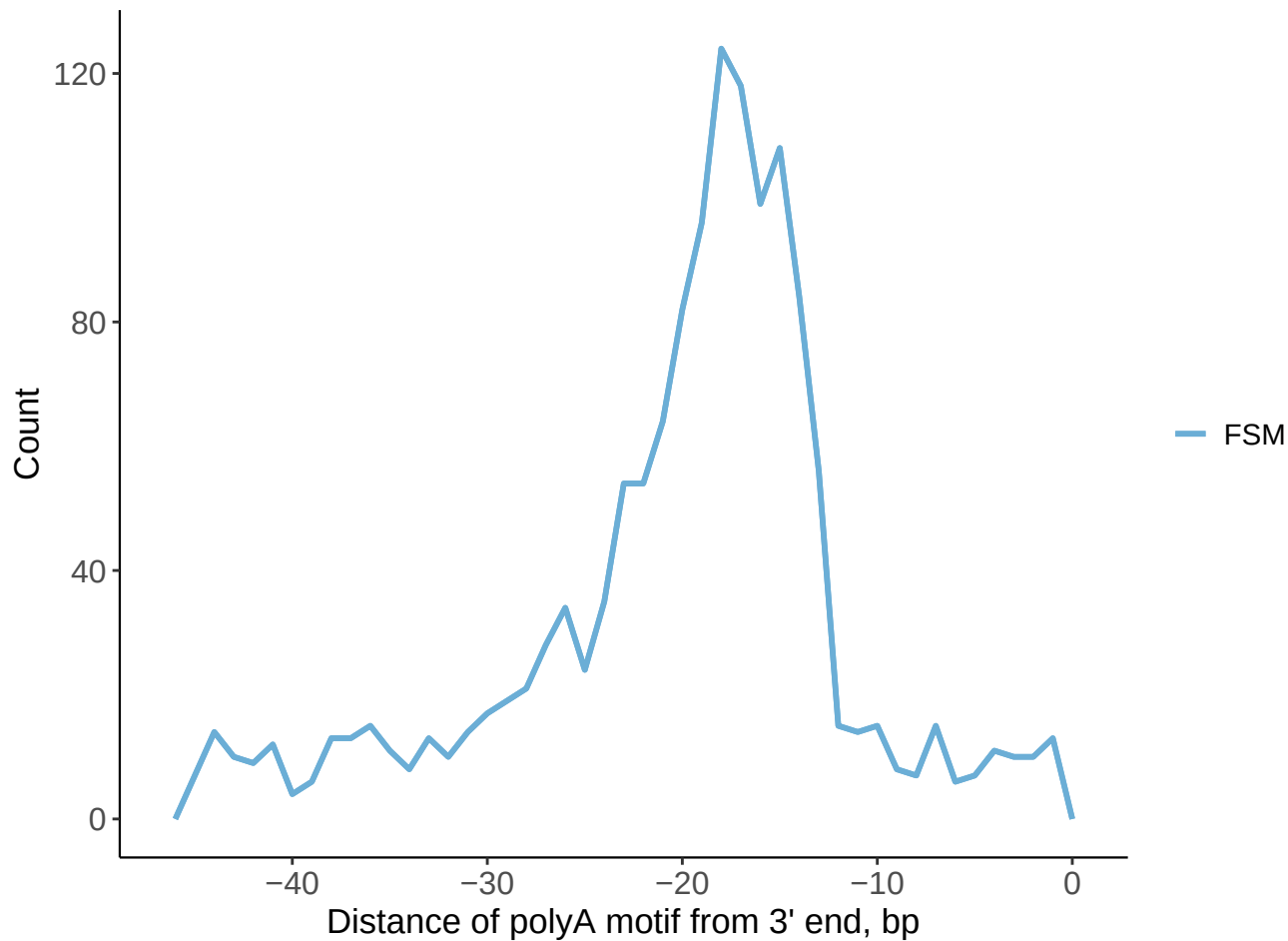
Distance to annotated Transcription Termination Site (TTS) FSM Reference Match

Negative values indicate upstream of annotated termination site



PolyA Distance Analysis

Distance of Detected PolyA Motif From 3' end



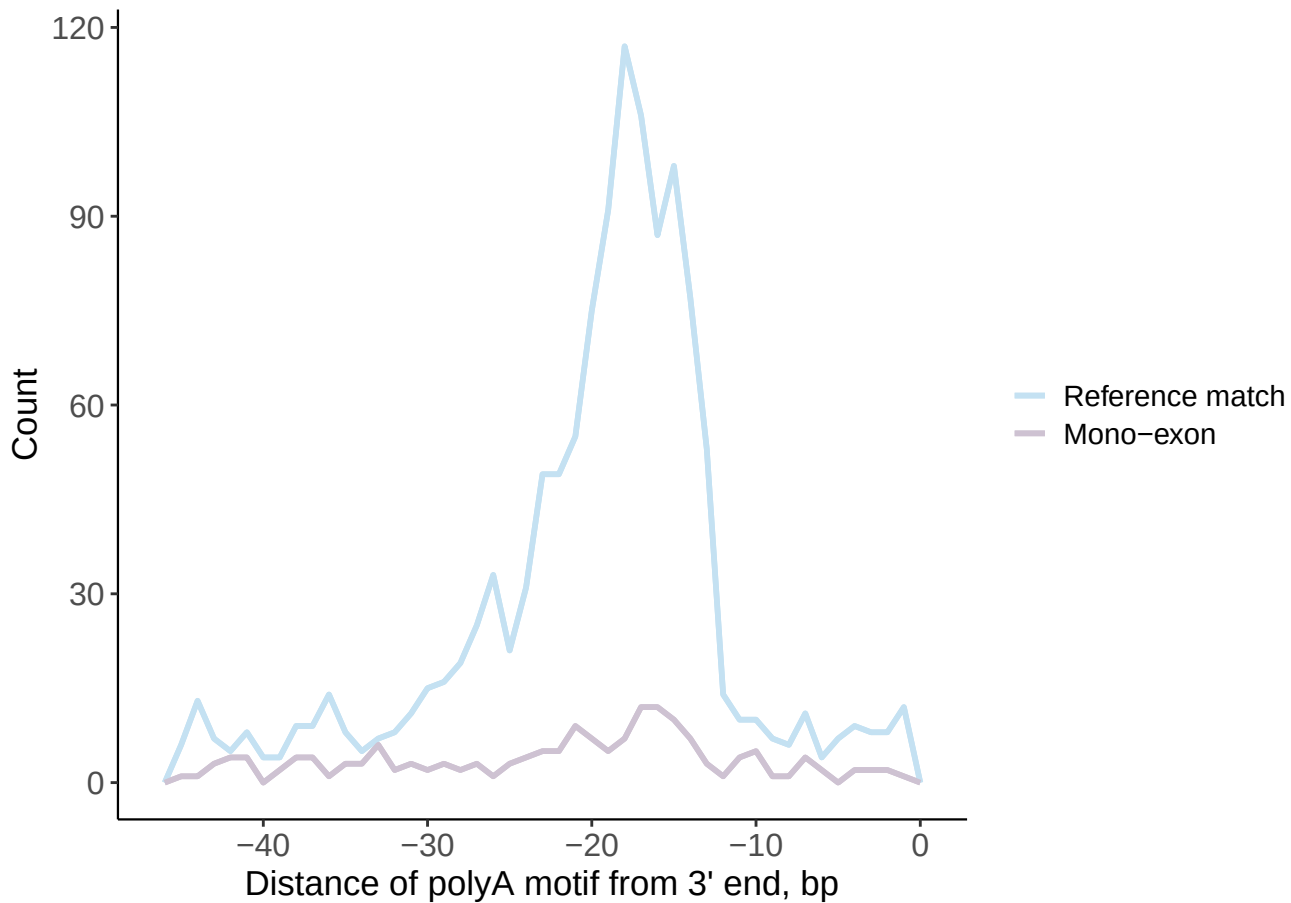
Number of polyA Motifs Detected

Category	Count	polyA Detected	%
FSM	2428	1407	58

Frequency of PolyA Motifs

Motif	Count	%
AATAAA	809	57.5
ATTAAA	200	14.2
AGTAAA	55	3.9
AAGAAA	43	3.1
TATAAA	39	2.8
GGGGCT	35	2.5
AATACA	33	2.3
GATAAA	33	2.3
TTTAAA	30	2.1
AAAAAG	29	2.1
CATAAA	23	1.6
AAAACA	21	1.5
ACTAAA	21	1.5
AATGAA	17	1.2
AATATA	11	0.8
AATAGA	8	0.6

Distance of Detected PolyA Motif From 3'End by FSM and ISM Subcategories



Distance of Detected PolyA Motif From 3'End by Non-FSM/ISM Subcategories

Count

Distance of polyA motif from 3' end, bp

Number of polyA Motifs Detected

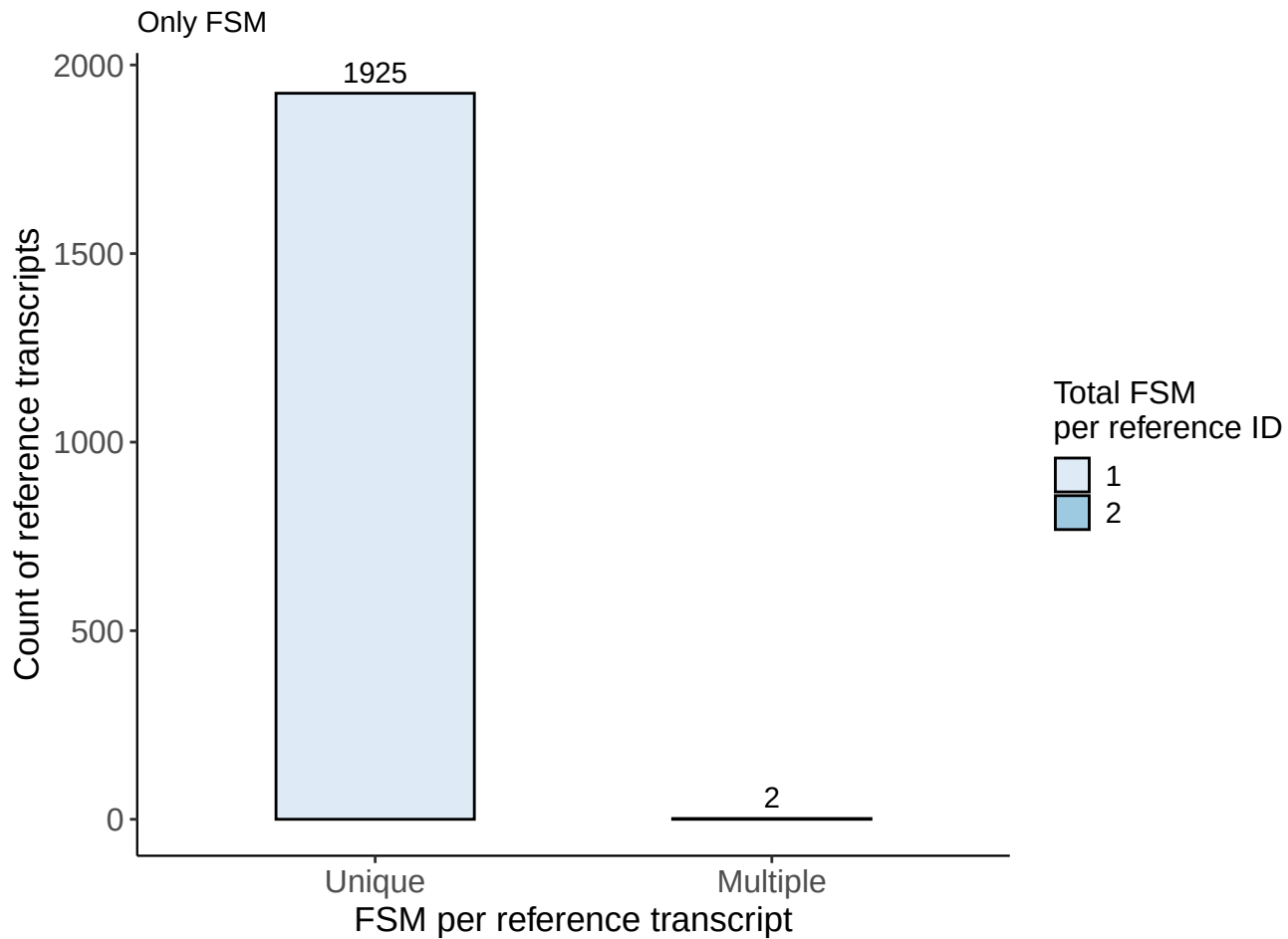
Subcategory	Count	polyA Detected	%
Reference match	1929	1241	64
Mono-exon	499	166	33

Frequency of PolyA Motifs

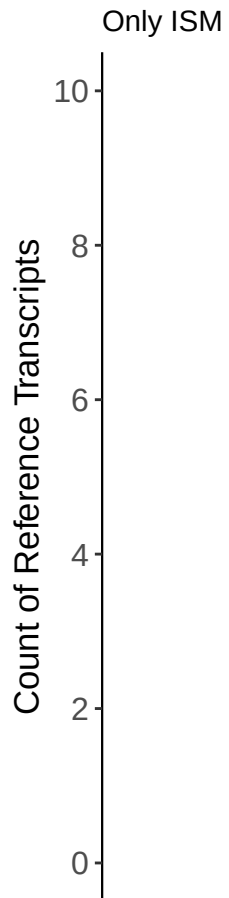
Motif	Count	%
AATAAA	809	57.5
ATTAAA	200	14.2
AGTAAA	55	3.9
AAGAAA	43	3.1
TATAAA	39	2.8
GGGGCT	35	2.5
AATACA	33	2.3
GATAAA	33	2.3
TTTAAA	30	2.1
AAAAAG	29	2.1
CATAAA	23	1.6
AAAACA	21	1.5
ACTAAA	21	1.5
AATGAA	17	1.2
AATATA	11	0.8
AATAGA	8	0.6

Redundancy Analysis

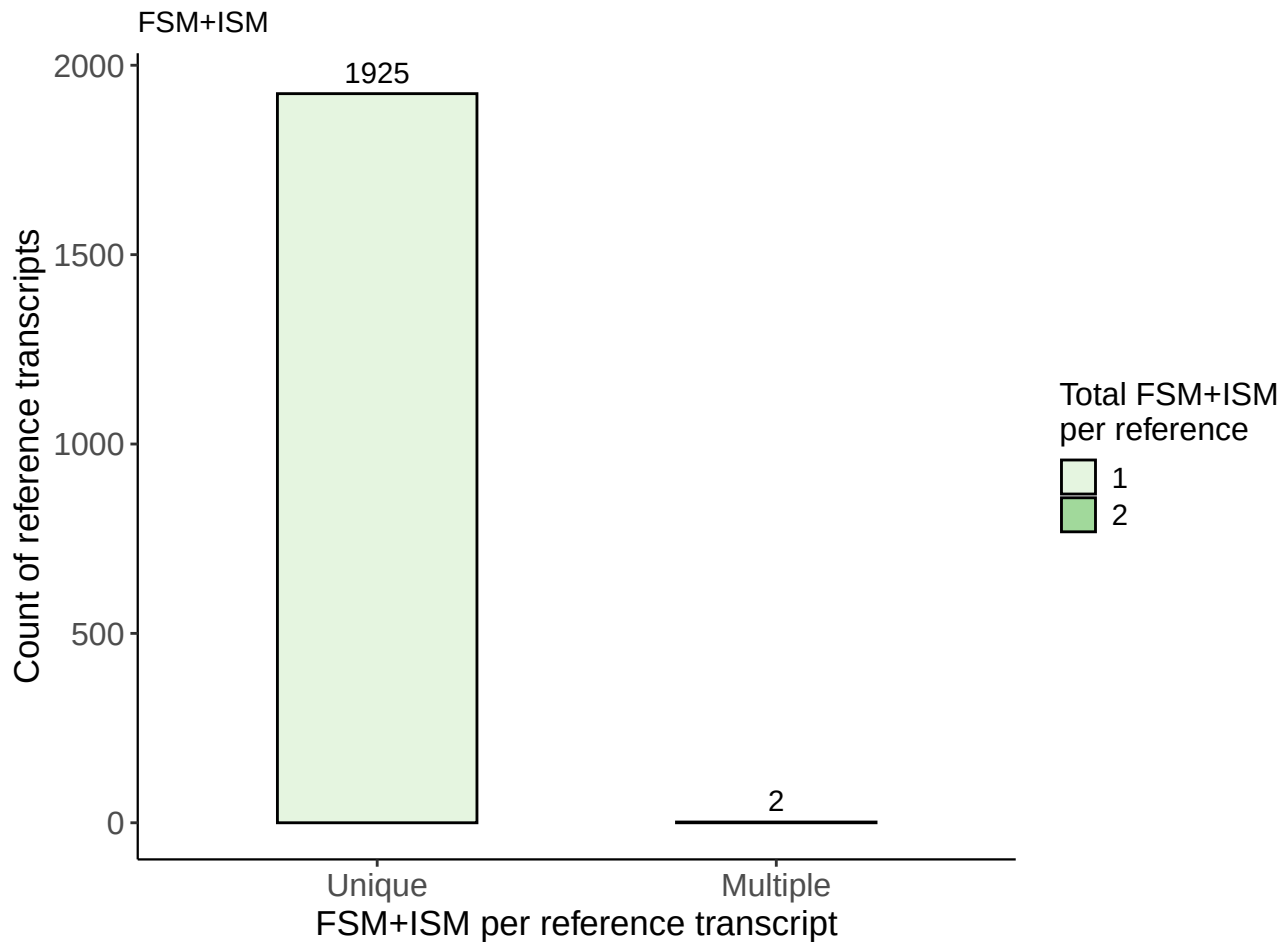
Reference Transcript Redundancy



Reference Transcript Redundancy

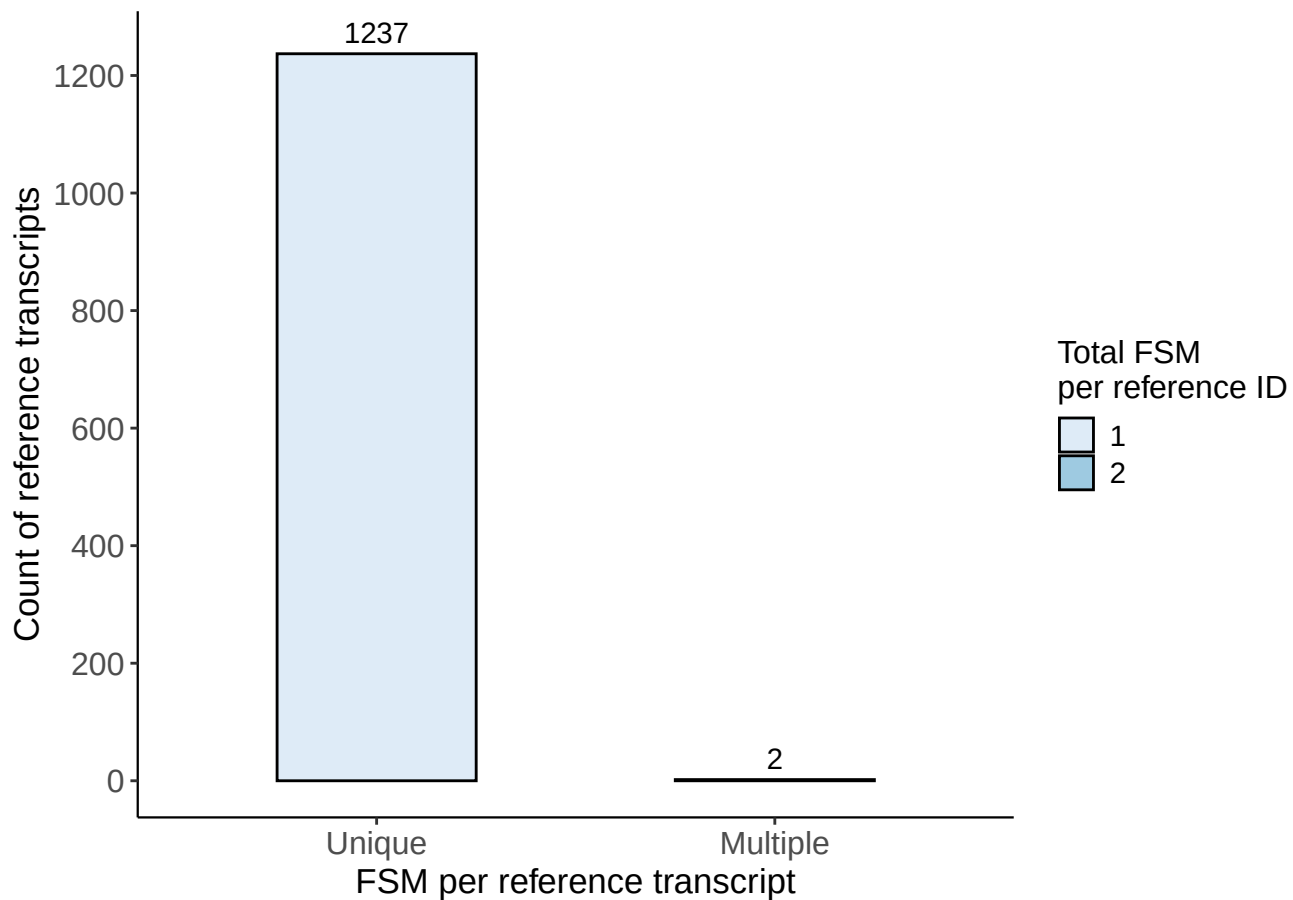


Reference Transcript Redundancy



Reference Transcript Redundancy

Only FSM with a polyA motif found



Reference Transcript Redundancy

Only ISM with a polyA motif found

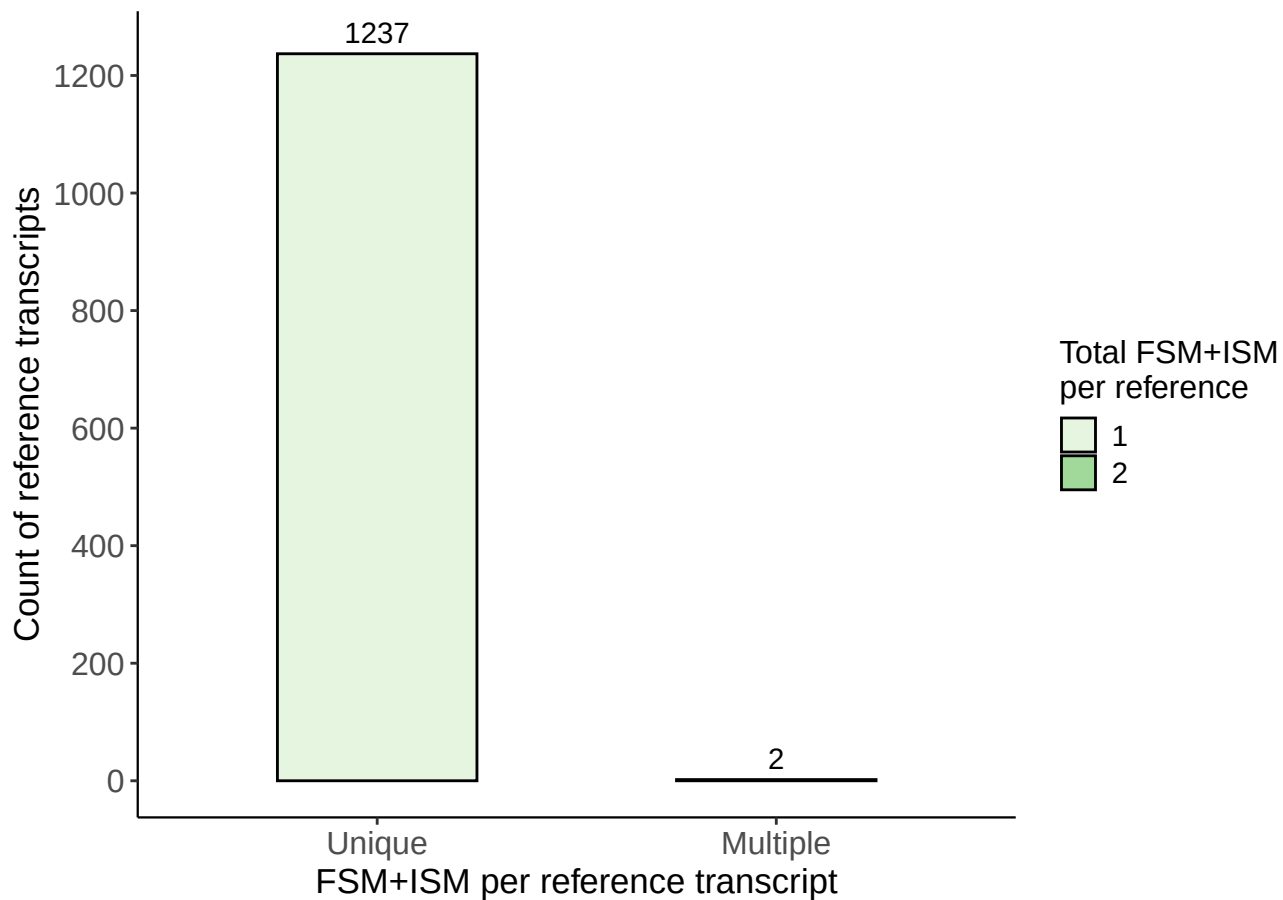
Count of reference transcripts

10
8
6
4
2
0

ISM per reference transcript

Reference Transcript Redundancy

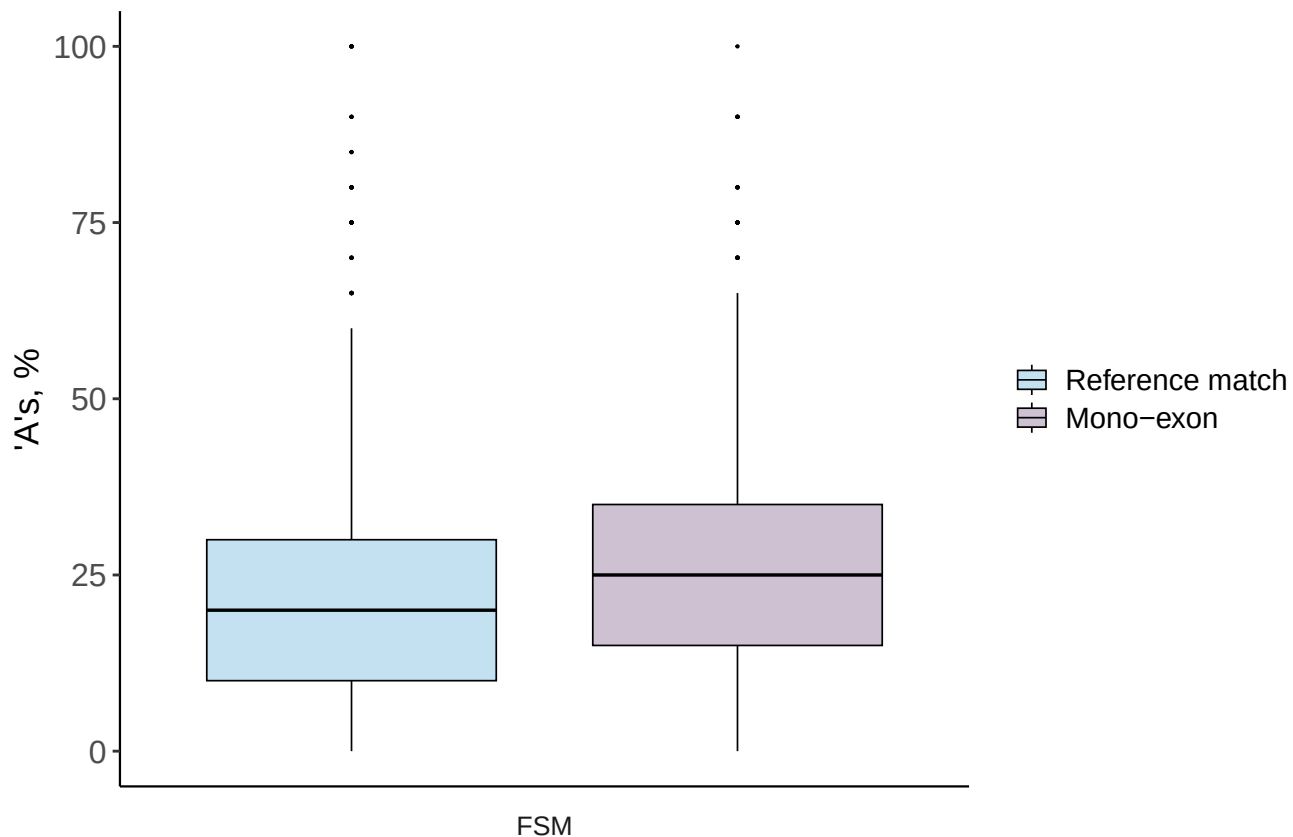
FSM+ISM with a polyA motif found



Intra-Priming Quality Check

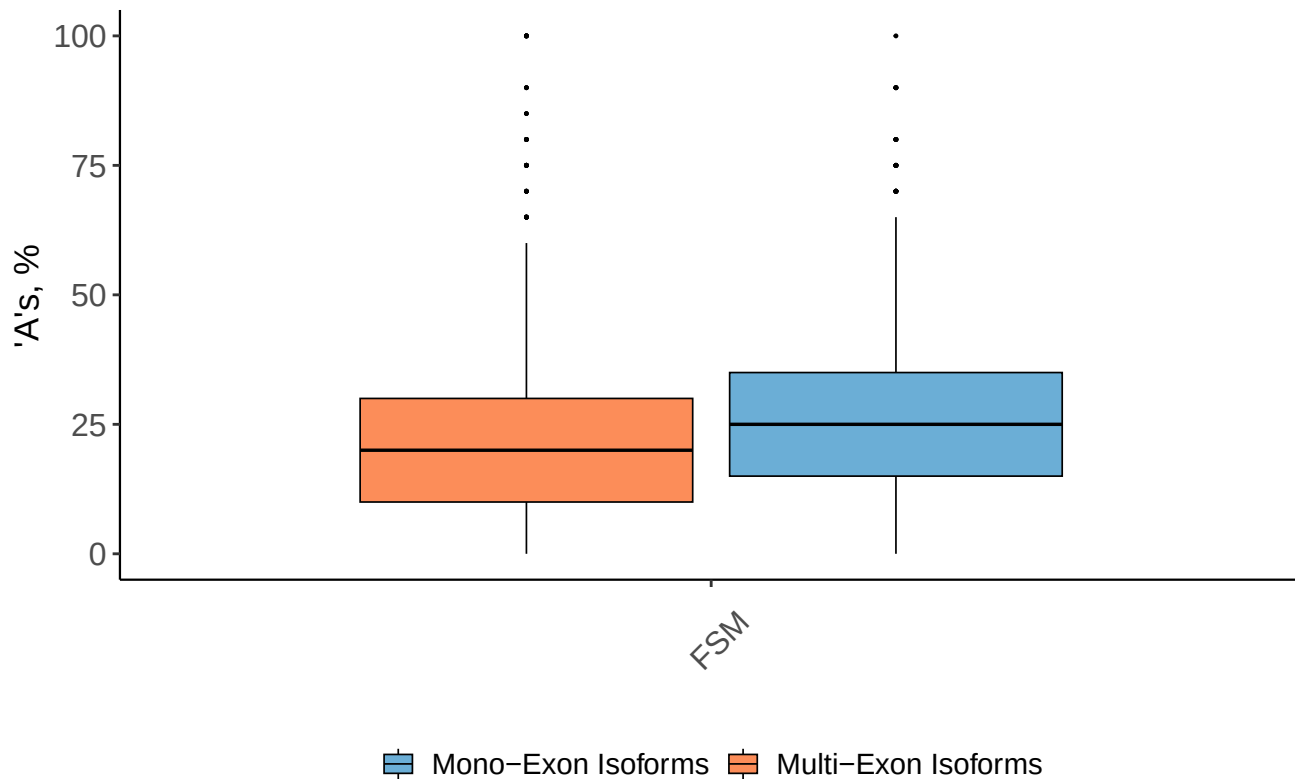
Possible Intra-Priming by Structural Category

Percent of genomic 'A's in downstream 20 bp



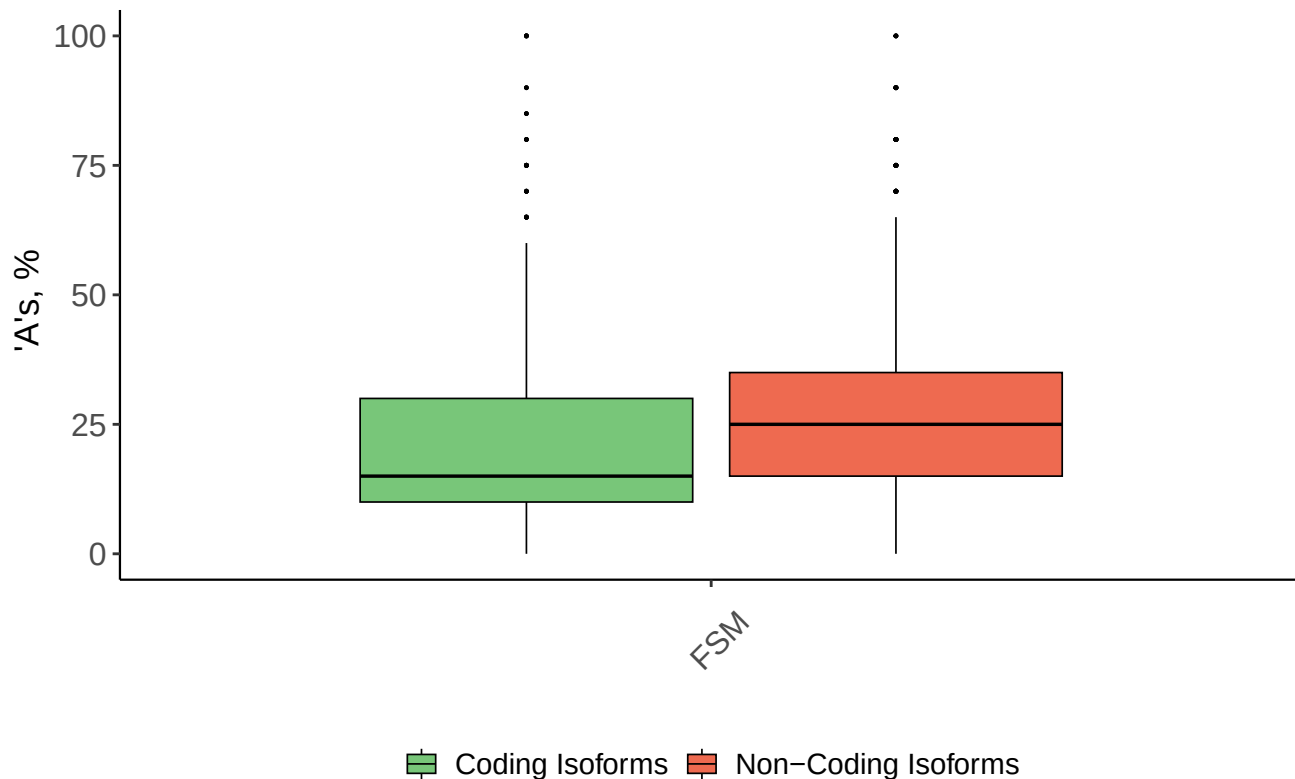
Mono- vs Multi-Exon Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp



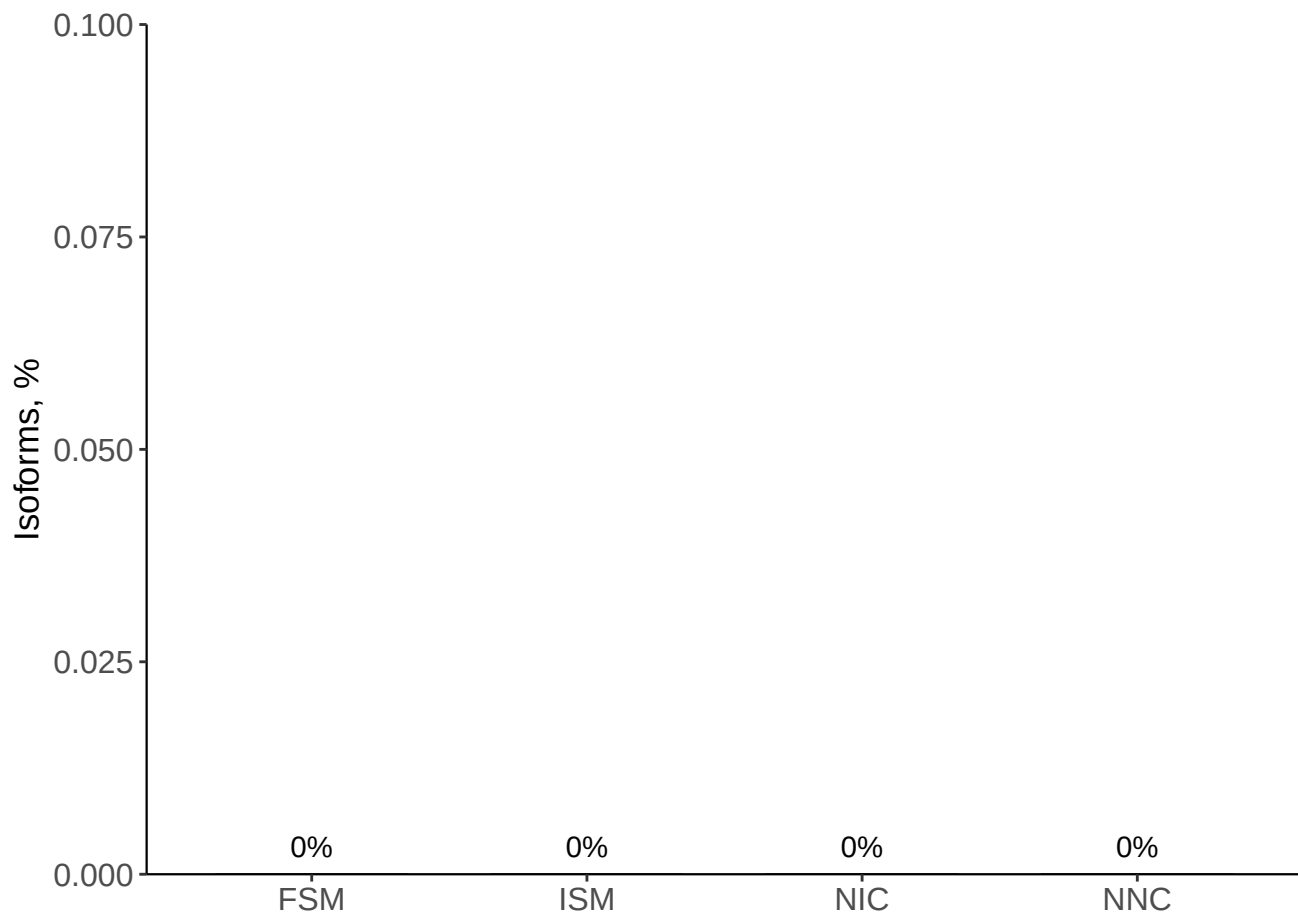
Coding vs Non-Coding Possible Intra-Priming

Percent of genomic 'A's in downstream 20 bp

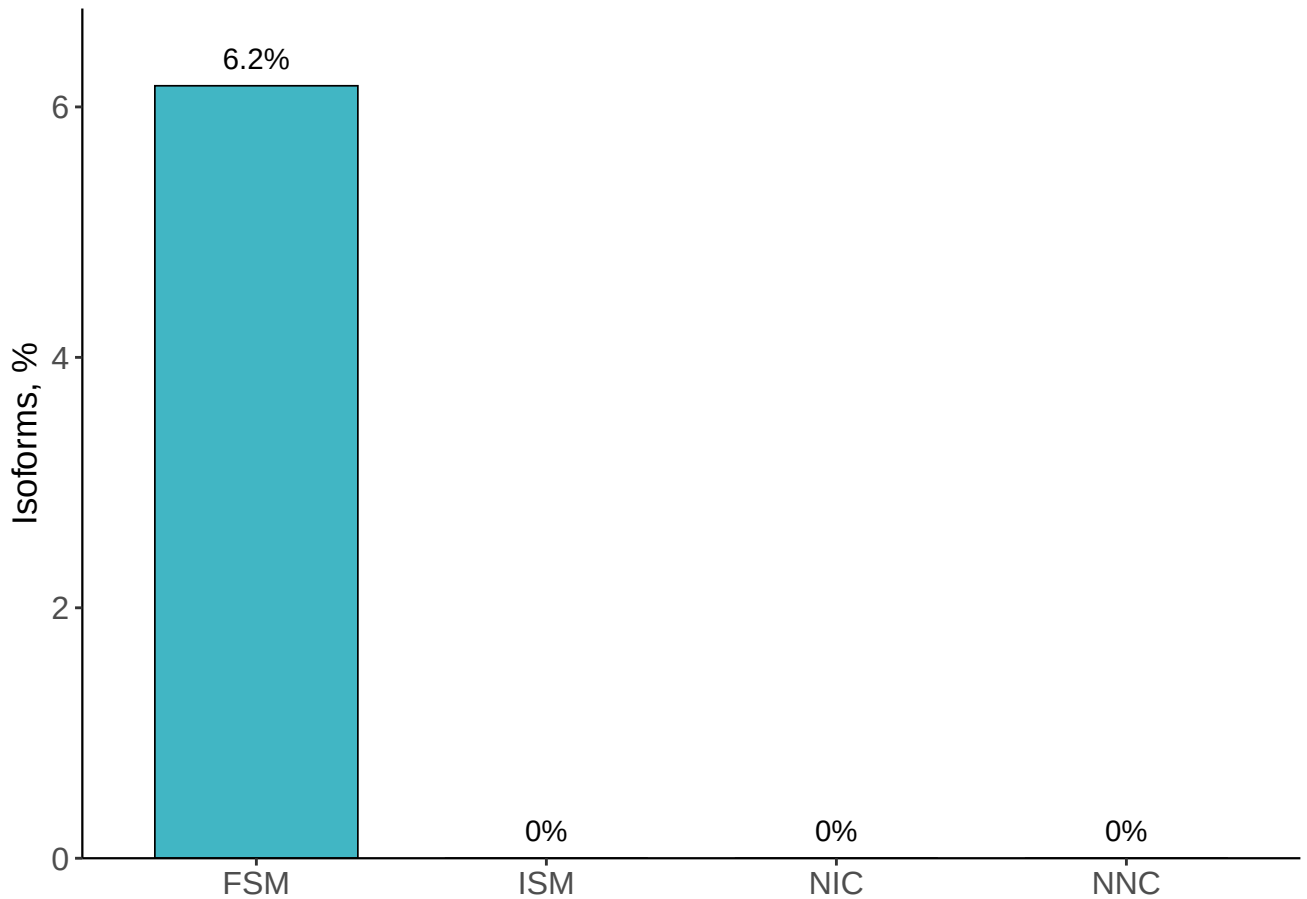


Features of Bad Quality

RT-switching

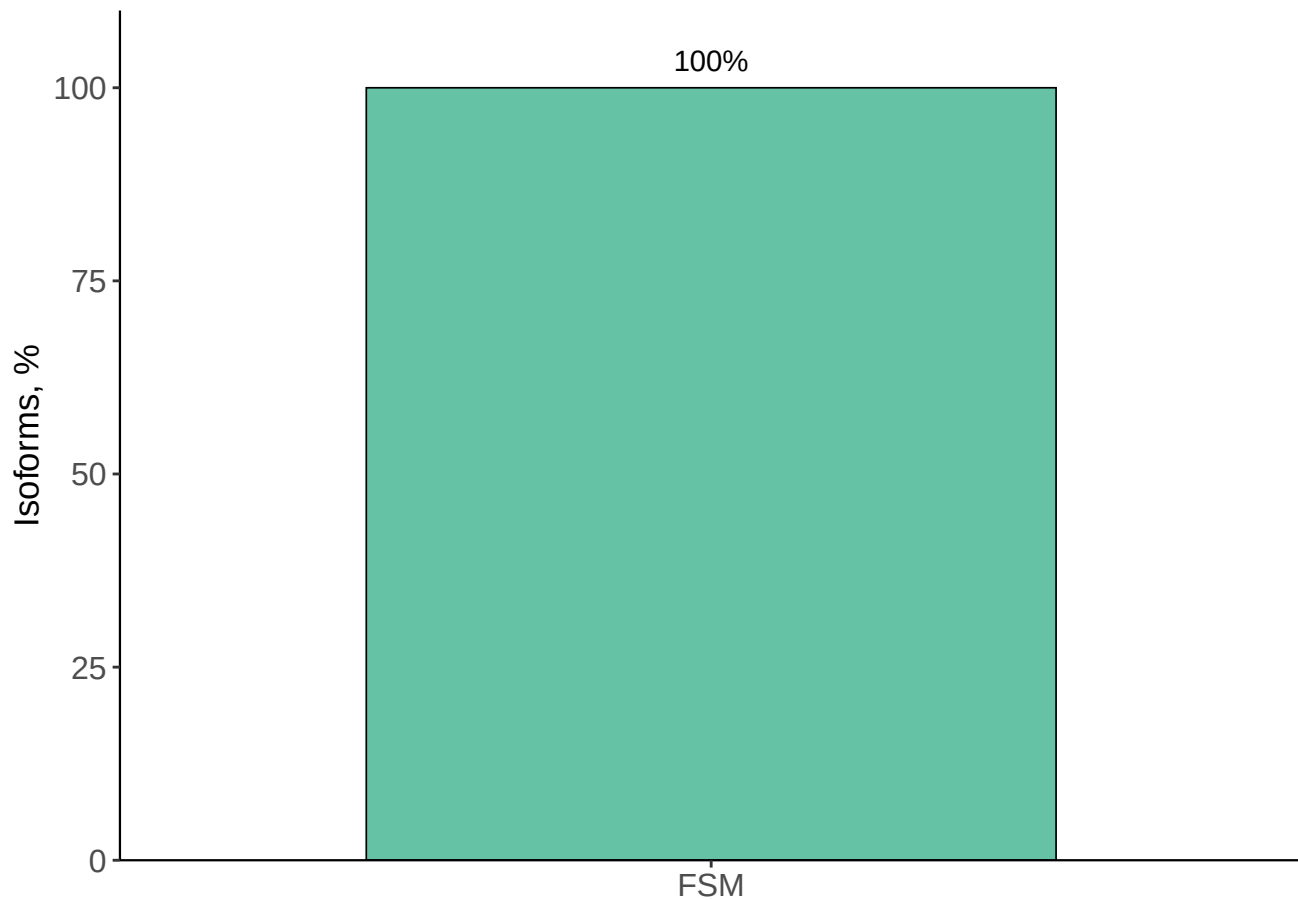


Non-Canonical Junctions

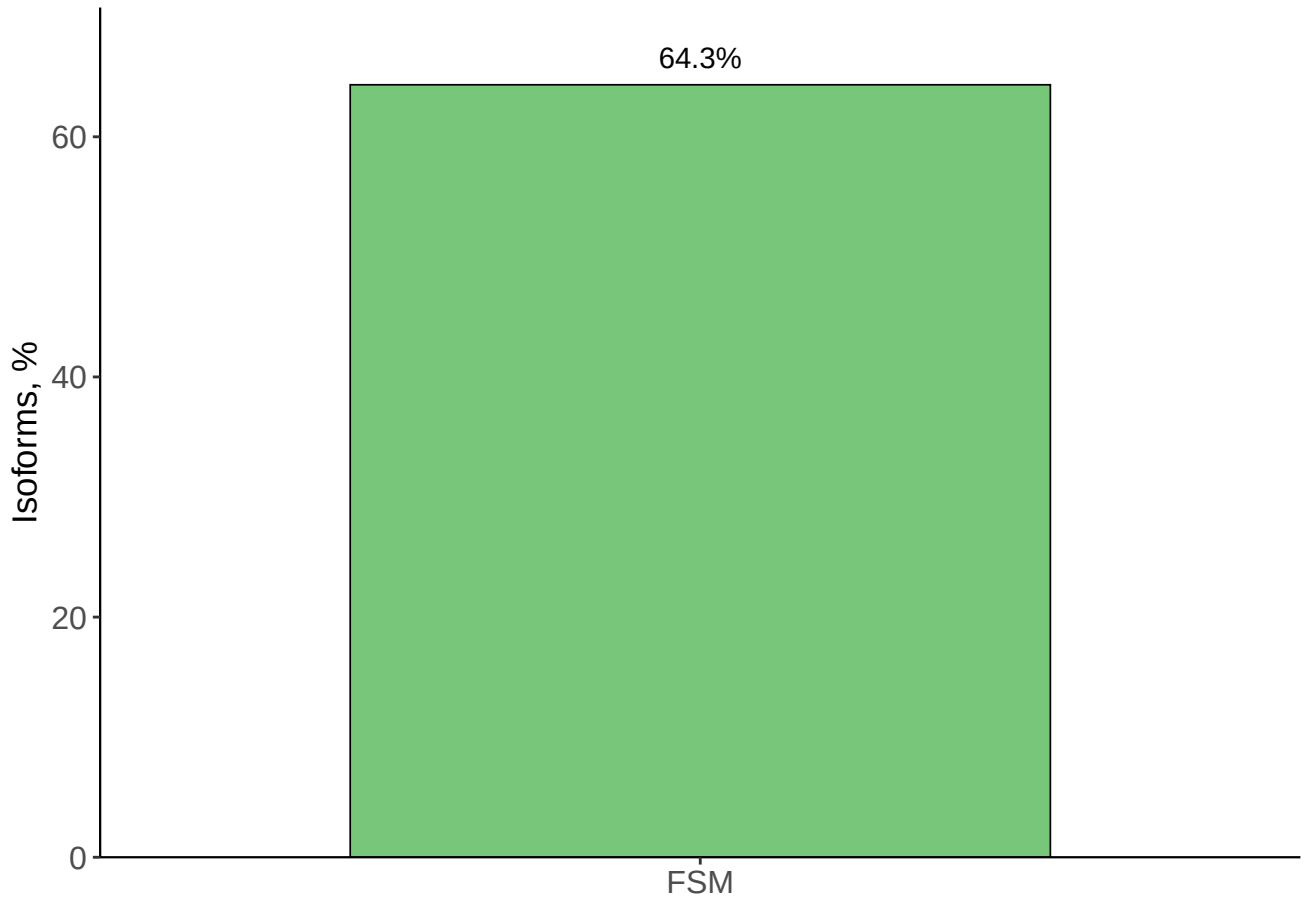


Features of Good Quality

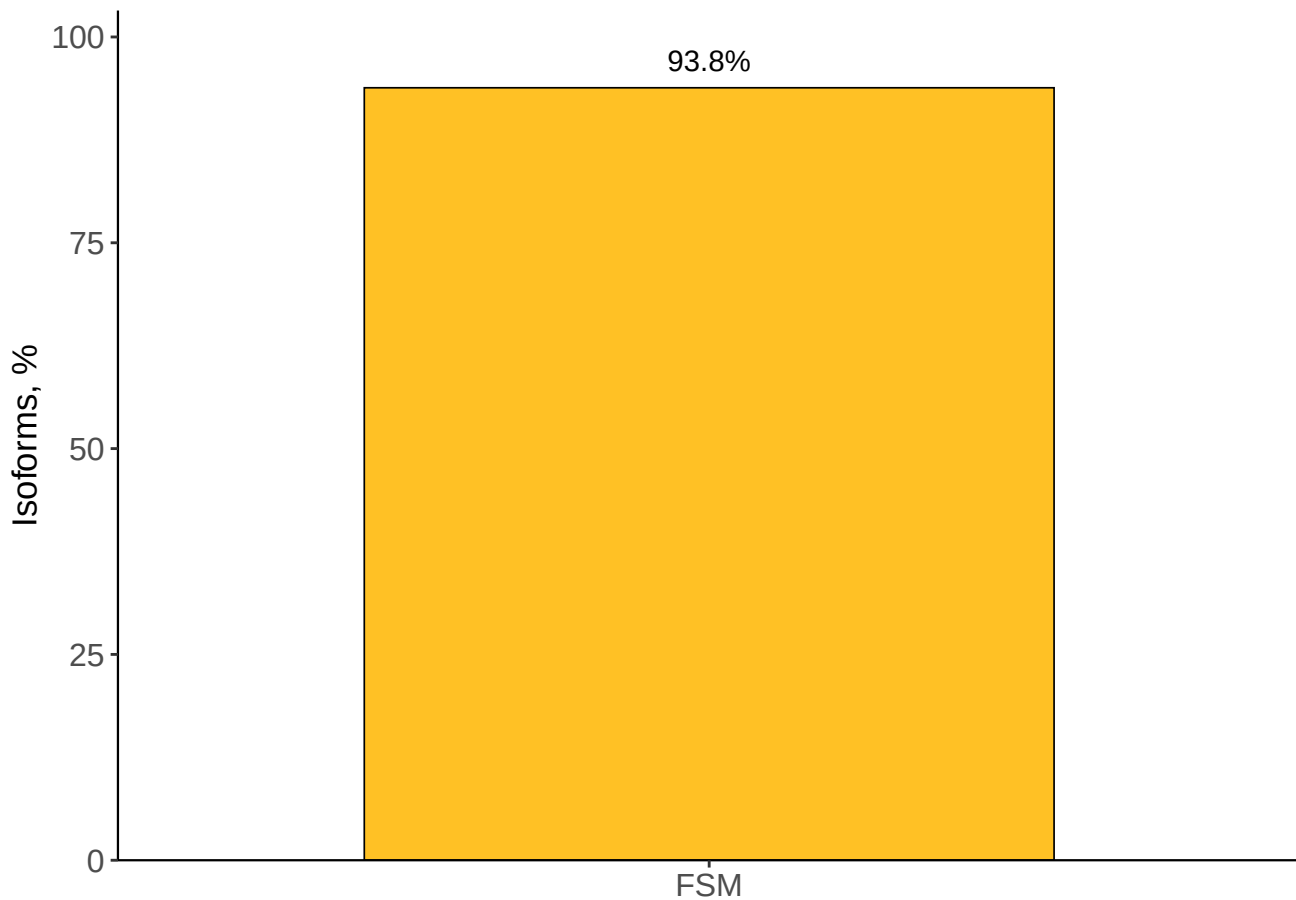
Annotation Support



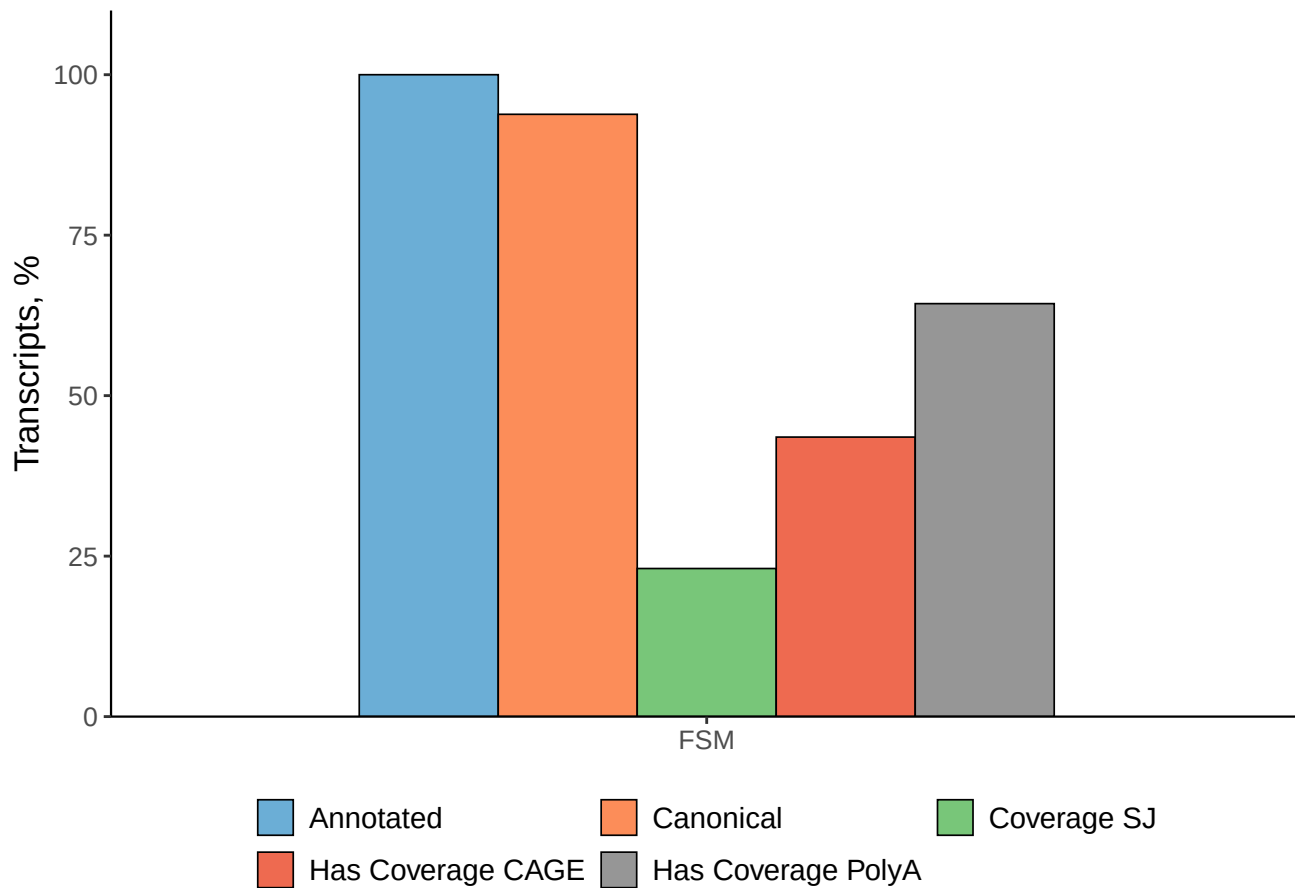
PolyA Support



All Canonical Junctions

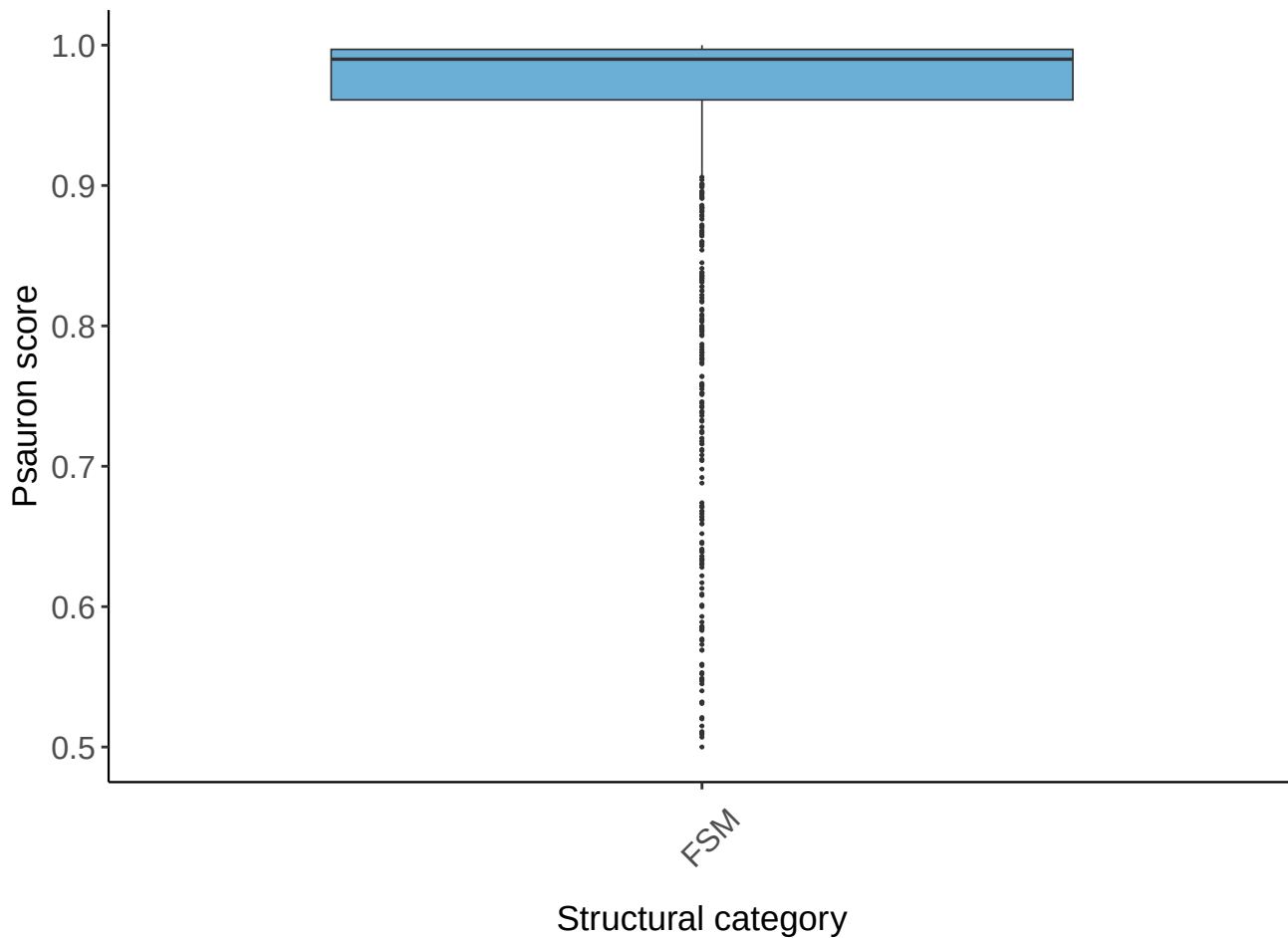


Good Quality Control Attributes Across Structural Categories

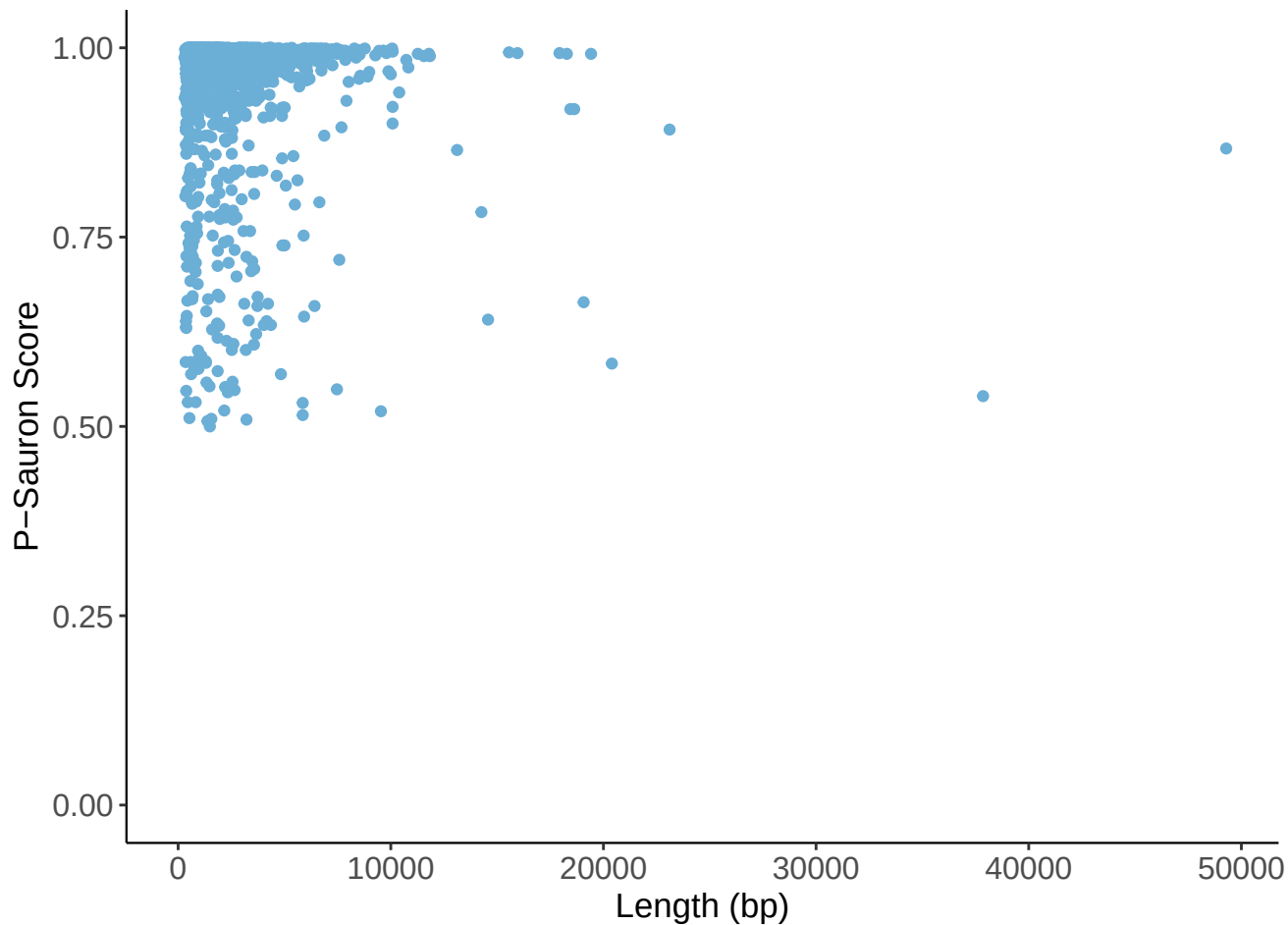


TransDecoder2 CDS prediction statistics

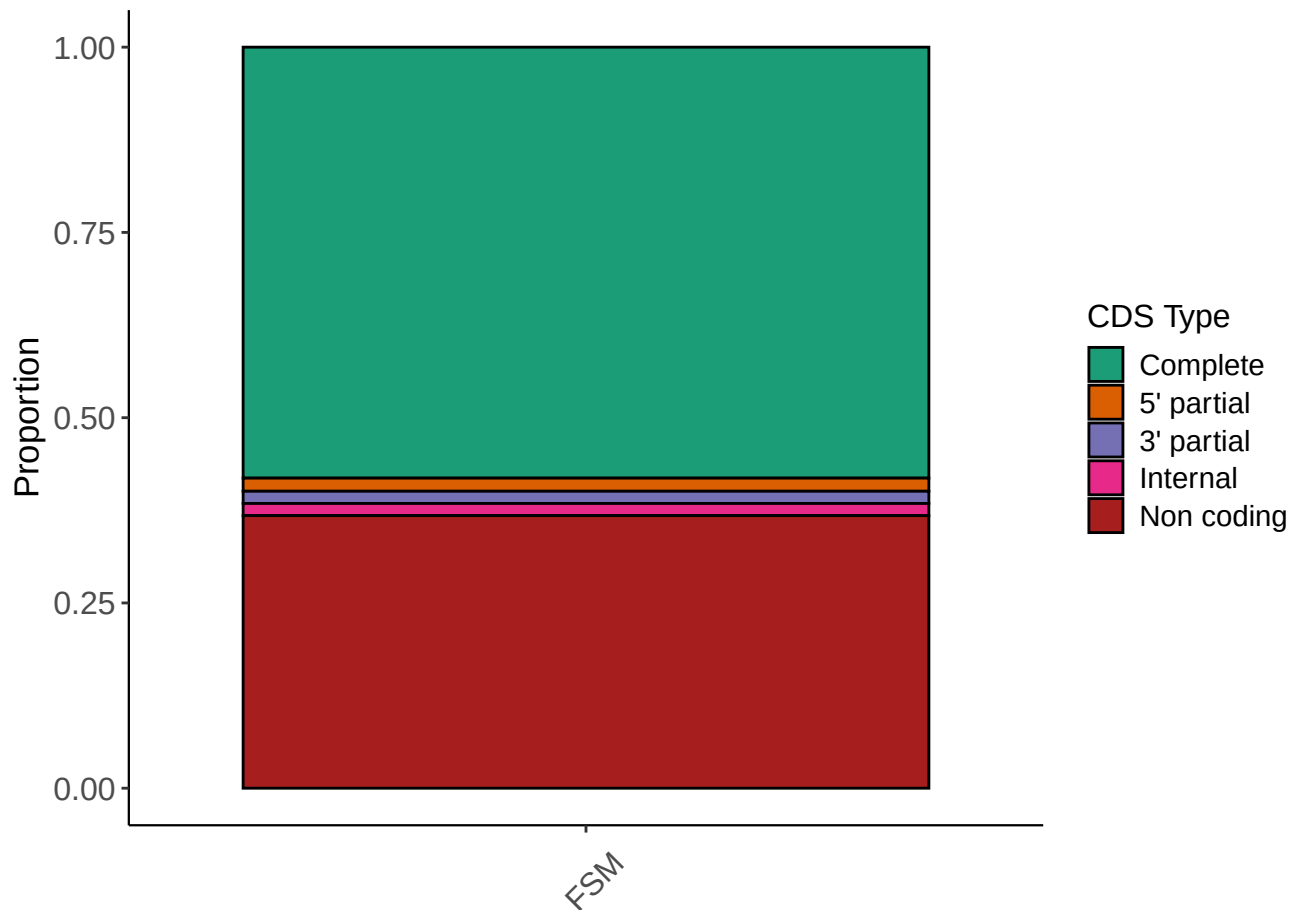
Psaaron score per structural category



P-Sauron Score vs Length (FSM)



CDS type by subcategory



Psauron score per ORF type

